

A Generalized Graph-Based Cluster Dynamics Framework for Irradiated Materials

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Abstract

We present a modular, graph-based cluster-dynamics (CD) framework for the long-term evolution of radiation-induced defect populations in crystalline solids and demonstrate it on the ferritic –martensitic steel EUROFER-97 under fission and fusion irradiation. Two algorithmic primitives carry the formulation: a polarity index $\sigma \in \{-1, +1\}$ that splits the state space into vacancy- and SIA-type sub-spaces without a host-specific taxonomy, and a reaction admissibility graph (RAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ whose edges encode every elementary reaction with stoichiometric tuples and rate kernels. The master equation reduces to the graph-walking accumulator $dc_a/dt = G_a + \sum_r S_{ar} J_r(\mathbf{c}) - D_a c_a$ applied edge-by-edge over $\mathcal{E}$. Specialization to EUROFER-97 yields size-resolved equations for SIA clusters, vacancy/helium cavities, and free helium, together with exact Frenkel-pair and helium-conservation laws and two helium-occupancy reductions (a mean-field loading $\bar{\ell}(m)$ for fusion-relevant He/dpa ratios and a decoupled scalar inventory Q_{tot} for fission). A logarithmic size-bin moment method with piecewise-constant and linear-hat Galerkin closures cuts the cluster-size ODE count by an order of magnitude. The bordered-banded CD Jacobian is solved with a tailored Sherman–Morrison–Woodbury–preconditioned GMRES core, all packaged in the open-source RADCLUSTER code. Demonstration runs spanning 10^{-6} to 100 dpa—with more than 20,000 ODEs on a single workstation—reproduce TEM loop and cavity sizes for EUROFER-97 without parameter tuning, identify the SIA-mobility cutoff i_{mobile} as the dominant phase-space knob (an order-of-magnitude loop-density swing), and show that the linear-hat bin-

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moment reduction delivers a $\sim 40\times$ speed-up over the fully discrete reference while matching every macroscopic target within a few percent.

Keywords: cluster dynamics, radiation damage, reaction admissibility graph, ferritic-martensitic steel, helium-vacancy clusters, size-bin moments

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1. Introduction

Structural materials in fission and fusion energy systems must operate under prolonged exposure to neutron radiation that displaces lattice atoms at rates of 10^{-7} to 10^{-5} dpa/s and accumulates damage to total doses of 10–200 dpa. The microstructural changes that result—nucleation and growth of vacancy and self-interstitial atom (SIA) clusters, void and bubble formation, dislocation-loop populations, irradiation-induced segregation, and precipitate coarsening or dissolution—are the principal sources of property degradation, manifesting as hardening, embrittlement, swelling, irradiation creep, and shifts in the ductile-to-brittle transition temperature. Predicting these property changes from first principles, *over the relevant time and length scales* and across the full range of temperature, dose, dose rate, and helium-to-dpa ratio, is the central quantitative challenge of radiation materials science.

The cluster-dynamics (CD) framework, introduced for radiation damage problems during the late 1960s and early 1970s [1–4], remains the workhorse mesoscopic theory for this challenge. The earliest treatment of colloid clustering by Chandrasekhar in the 1940’s [5] (Eqn. 447) paved the way for CD formulations of defects as reacting particles [1]. CD treats each defect species and size as a separate population whose concentration evolves under a coupled set of master rate equations balancing in-cascade production, diffusion-controlled reactions, thermal emission, and absorption at fixed sinks [6–8]. Its three defining advantages are direct access to time scales (10^{-6} – 10^8 s) that lie far beyond molecular dynamics [9, 10], explicit resolution of the cluster-size distribution, and therefore of every quantity that enters macroscopic property models—from sink-strength models to swelling rates [11, 12]—and modest computational cost relative to spatially resolved kinetic Monte Carlo or atomistic simulation. Together, these features make CD the natural meeting point for atomistic-scale parameterization and engineering-scale prediction [13–15]. Since its inception, CD has been deployed across an unusually broad materials portfolio: void swelling and helium effects on cavity stability in austenitic stainless steels and Fe–Cr–Ni model alloys [3, 16, 17]; SIA-cluster and Cr-precipitate kinetics in ferritic and ferritic–martensitic steels [18–20]; helium and hydrogen-isotope retention in plasma-facing tungsten and W–Re alloys [21–23]; evolution and bubble nucleation in nuclear fuels [24]; and the $\langle a \rangle$ -loop and $\langle c \rangle$ -loop populations that underlie irradiation growth

The several review articles that have shaped the field [15, 25–28] never-

theless converge on a common diagnosis: while CD is conceptually mature, the practical landscape is fragmented. Existing codes specialize in a single host material, in a particular reaction subset, or in a particular solver philosophy, and there is no widely shared, transparent, and modular formulation that exposes *which* physics has been included, *which* approximation has been invoked, and *which* parameter has been calibrated. This fragmentation has produced several persistent gaps. The first is a lack of host independence at the level of the algorithmic core: specializing the same machinery for different materials typically requires a non-trivial rewrite of the solver, even though the underlying mathematical structure remains unchanged. The second is the state-space management of helium–vacancy clusters: tracking $\text{He}_\ell V_m$ explicitly across the full (ℓ, m) grid is prohibitive [13, 29, 30], and reduced descriptions that track only a mean helium loading per cavity have been used heuristically [31, 32] without making explicit the limits in which they are exact, the leading correction terms, or the conservation laws they preserve. The third is the closure of the cluster-size distribution: Fokker–Planck representations [31, 33, 34] lose accuracy at the discrete-to-continuous transition, grouping or binning closures [11, 15, 35, 36] introduce mass-conservation errors unless the inter-bin flux is explicitly designed, and few formulations rigorously analyze closure accuracy in a way that lets the user trade fidelity for system size with quantified error. The fourth is stiffness and solver design: the CD master equations carry both narrow-banded structure (within a single species) and dense bordering blocks (cross-species coupling, sink strengths, conservation constraints), generic ODE solvers and standard preconditioners perform poorly on this structure, and few existing CD codes invest in tailored numerical strategies that exploit the bordered-banded form or document explicit conservation-law diagnostics as a built-in part of the solver [37].

The present paper addresses these four gaps in a modular framework whose components can be selected, replaced, or specialized depending on the host material and the irradiation environment. The goal is to deliver a general, transparent CD formulation in which every reaction rate, every binding energy, and every closure approximation is traceable to its physical origin and its calibration data, and in which the formulation specializes in a particular host material—taking the reduced-activation ferritic–martensitic steel EUROFER-97 as the principal example—without modifying the algorithmic core. The framework provides two tailored helium–vacancy reductions derived as systematic limits of the same parent master equations:

a fast-helium equilibration limit appropriate to fusion-relevant He/dpa ratios, which tracks the mean helium loading per cavity, and a decoupled helium inventory limit appropriate to fission spectra, which tracks a single scalar bubble-helium inventory. Both preserve the Frenkel-pair and helium conservation moments exactly. State-space reduction is further developed through a logarithmic size-bin moment method with two complementary closures (piecewise-constant and Galerkin shape-functions), each accompanied by an explicit accuracy and mass-conservation analysis so that system size can be traded against fidelity in a controlled way. The numerical implementation rests on a tailored Sherman–Morrison–Woodbury preconditioner that exploits the bordered-banded Jacobian [38–40], together with conservation-law diagnostics used during time integration; the complete formulation is realized in the open-source RADCLUSTER code.

The framework is host-independent, thus facilitating future research. Adding a new host material is a declaration, not a refactor: it amounts to specifying the admissible population set $\mathcal{P}_\sigma$, the admissible vertex set $\mathcal{V}$, the admissible edge set $\mathcal{E}$, and the rate-kernel library $\mathbf{k}$, while the solver, the preconditioner, the state-space reductions, and the conservation diagnostics are inherited unchanged. The state-space reduction theorems are themselves graph-theoretic; both are stated and proved on the abstract graph and inherited by any host. The analogy to object-oriented programming is precise: the abstract graph $\mathcal{G}$, together with its solver, preconditioner, and reduction theorems, acts as an abstract base class that defines the interface and implements all methods; and running a simulation for a particular irradiation scenario is instantiation. New materials enter the framework as subclasses, not rewrites.

Section 2.1 defines the polarity-indexed cluster state space and the composition vector that distinguishes host coordinates from solute coordinates. Section 2.2 defines the reaction admissibility graph and catalogs the elementary edge classes—growth, shrinkage, dissociation, recombination, solute trapping, inter-population transfer, coalescence, source, and sink—from which any specialized application selects its subset. Section 2.3 develops the hierarchical decomposition of the graph by polarity and by population, and shows that the bordered-banded sparsity of the master-equation Jacobian is a direct consequence of that decomposition. Section 2.4 illustrates the modularity by sketching the host-specific declarations for ferritic–martensitic steels. Section 3 casts the master equation in a code-generation-ready form, comprising stoichiometric matrices and conservation laws.

The abstract framework is then instantiated for ferritic-martensitic steels.

Section 4 works out the full instantiation of the RAG for EUROFER-97: the reaction admissibility graph, the master equations for SIA clusters, vacancy/helium cavities, and free helium; the Frenkel-pair and helium-conservation laws; the swelling identity; the two helium-occupancy reduction modes; and the explicit mapping to RADCLUSTER code variables [41]. The helium-vacancy state-space reductions are derived in Section 5 as systematic limits of the master equations, and the logarithmic size-bin moment reduction is developed in Section 5.4 together with both closures and the accompanying accuracy and conservation analysis. Demonstration simulation results for EUROFER-97 under prototypical reactor conditions are presented in Section 6.

2. Reaction Network Graphs

2.1. Indexed Cluster State Space

A cluster identifier is a four-tuple

$$\boldsymbol{\tau} = (\sigma, n, p, \mathbf{c}) \in \{-1, +1\} \times \mathbb{Z}_{\geq 1} \times \mathcal{P} \times \mathcal{C}, \quad (1)$$

read as “a cluster of polarity σ , size n , located in the population p , carrying solute composition $\mathbf{c}$.” The polarity index $\sigma \in \{-1, +1\} = \{-1, +1\}$ classifies the cluster by the sign of its net Frenkel-pair content with respect to the host lattice, being -1 for a vacancy-type and +1 for an interstitial-type. It is unaffected by solute trapping: a helium–vacancy cluster $\text{He}_j V_n$ retains $\sigma = -1$ regardless of j .

The size $n \in \mathbb{Z}_{\geq 1}$ is the count of host Frenkel-pair sites of the polarity in question carried by the cluster: $n = n_{\text{vac}}$ for $\sigma = -1$ and $n = n_{\text{sia}}$ for $\sigma = +1$. In particular, the monomers are $(\sigma = -1, n = 1) = V$ and $(\sigma = +1, n = 1) = I$. Rate kernels are written as $k_{\sigma\sigma'}(n, n')$ and addressed by polarity, and the state arrays are indexed on a non-negative size axis.

The composition vector $\mathbf{c} = (c_1, \dots, c_S) \in \mathbb{Z}_{\geq 0}^S = \mathcal{C}$ records the count of each non-host species $s = 1, \dots, S$ trapped in the cluster. For the EUROFER-97 application, $S = 2$ with $\mathbf{c} = (c_{\text{He}}, c_{\text{H}})$. For a tungsten plasma-facing application, S might be 3 to include implanted deuterium. The composition vector is independent of σ and n . Its admissible range may depend on (σ, n, p) through host-specific saturation rules (e.g. a helium bubble saturates at a He/V ratio set by self-trapping energetics), and these rules enter as edge admissibility constraints on the graph of Section 2.2.

The population label $p \in \mathcal{P}$ partitions clusters of the same polarity, size, and composition into spatially or microstructurally distinct populations, between which transport is itself a CD reaction. Typical population sets are

$$\mathcal{P}_- = \{ \text{bulk, GB, disl, ppt} \} \quad (\text{vacancy clusters}), \quad (2)$$

$$\mathcal{P}_+ = \{ \text{bulk-}\langle 111 \rangle, \text{bulk-}\langle 100 \rangle, \text{disl-trapped} \} \quad (\text{bcc SIA clusters}), \quad (3)$$

$$\mathcal{P}_+^{\text{hep}} = \{ \text{bulk-}\langle a \rangle, \text{bulk-}\langle c \rangle, \text{basal-trapped, prismatic-trapped} \} \quad (4)$$

The set $\mathcal{P}$ is host-specific. Populations are not mere spatial coordinates; they are physically distinct phases of the cluster (different binding energies to surrounding microstructure, different mobility, different energetic stability).

The microstate of the system at time t is a function.

$$\mathbf{s}(t) : \mathcal{V} \rightarrow \mathbb{R}_{\geq 0}, \quad \boldsymbol{\tau} \mapsto C_{\boldsymbol{\tau}}(t), \quad (5)$$

where $C_{\boldsymbol{\tau}}(t)$ is the volumetric concentration of clusters of identifier $\boldsymbol{\tau}$, and $\mathcal{V}$ is the admissible vertex set of the reaction admissibility graph defined in Section 2.2. A particular host material with parameter set Θ specifies $\mathcal{V}$, $\mathcal{P}$, $\mathcal{C}$, and the saturation rules; the algorithmic core operates on $\mathbf{s}$ without further reference to Θ .

2.2. The Reaction Admissibility Graph

The reaction admissibility graph (RAG) is a labeled multidigraph.

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{\nu}, \mathbf{k}), \quad (6)$$

where $\mathcal{V} \subset \{-1, +1\} \times \mathbb{Z}_{\geq 1} \times \mathcal{P} \times \mathcal{C}$ is the (host-specific) vertex set of admissible cluster identifiers; $\mathcal{E} \subset \mathcal{V} \times \mathcal{V} \times \mathbf{N}$ is the edge set, where each edge $e = (\boldsymbol{\tau}_1, \boldsymbol{\tau}_2, \nu_e)$ carries a stoichiometric tuple $\nu_e \in \mathbf{N}$ that records all reactants and products of the elementary reaction; and $\mathbf{k} : \mathcal{E} \rightarrow \mathbb{R}_{\geq 0}$ assigns a rate kernel $k_e(t, \mathbf{s}, \Theta)$ to each edge, possibly $\mathbf{s}$ -dependent (sink-strength feedback, biased absorption) and parameter-dependent (binding energies, diffusivities).

The graph is host-specific in $(\mathcal{V}, \mathcal{E}, \boldsymbol{\nu})$ and calibration-specific in $\mathbf{k}$; the algorithmic core is the graph walker. Here, “graph walker” refers to the algorithmic traversal of the reaction admissibility graph. The ODE right-hand side is assembled by looping over admissible reaction edges $r \in \mathcal{E}$, evaluating the corresponding reaction flux $J_r(\mathbf{C})$, and accumulating its stoichiometric contribution $S_{ar} J_r$ to each affected cluster concentration.

Every edge in $\mathcal{G}$ falls into exactly one of the following classes, identified by the pattern of polarity and population at its endpoints and by the form of its stoichiometric tuple:

1. **Growth.** $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n+1, p, \mathbf{c})$ by absorption of a σ -monomer. Both endpoints share polarity, population, and composition.
2. **Shrinkage.** $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n-1, p, \mathbf{c})$ by absorption of a $-\sigma$ -monomer (impingement of the opposite polarity).
3. **Dissociation.** $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n-1, p, \mathbf{c})$ by thermal emission of a σ -monomer; the released monomer is added to the σ -monomer pool. The rate kernel involves a binding energy.
4. **Recombination (monomer cross-polarity edge).** $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n-1, p, \mathbf{c})$ paired with $(-\sigma, 1, p', \mathbf{0}) \rightarrow \emptyset$. A cluster of polarity σ and size n absorbs a single $(-\sigma)$ -monomer; the monomer is annihilated rather than added to any pool.
5. **Annihilation (cross-polarity binary hyperedge).** $(\sigma, n, p, \mathbf{c})$ and $(-\sigma, n', p', \mathbf{c}')$ jointly $\rightarrow (\sigma, |n-n'|, p_{\text{surv}}, \mathbf{c}_{\text{surv}})$, where

$$\sigma = \text{sgn}(n - n'), \quad \mathbf{c}_{\text{surv}} = \begin{cases} \mathbf{c} & n > n', \\ \mathbf{c}' & n < n', \\ \mathbf{0} & n = n'. \end{cases} \quad (7)$$

When $n = n'$ both precursor vertices are removed ($\rightarrow \emptyset$); when $n \neq n'$ the larger cluster survives at a reduced size with the polarity of the larger reactant and the composition of the surviving fragment. This is the opposite polarity analog of *coalescence*: it is a *hyperedge* with two precursor vertices and a *quadratic* rate kernel.

6. **Inter-population (cross-population edge).** $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n, p', \mathbf{c})$ with $p \neq p'$. This edge represents segregation, sweeping, or capture by a moving sink; it changes only the population coordinate.
7. **Solute trapping/detrapping (intra-vertex edge).** $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n, p, \mathbf{c} \pm \mathbf{e}_s)$, where $\mathbf{e}_s$ is the unit composition vector for species s . This edge changes only the composition coordinate.
8. **Coalescence (binary growth).** $(\sigma, n, p, \mathbf{c})$ and $(\sigma, n', p, \mathbf{c}')$ jointly $\rightarrow (\sigma, n+n', p, \mathbf{c} + \mathbf{c}')$. This is a hyperedge in the strict sense; in the implementation, it is represented as a multi-source edge with a quadratic rate kernel.

9. **Source (cascade-injection edge).** $\emptyset \rightarrow (\sigma, n, p, \mathbf{c})$ with a rate set by the displacement-cascade spectrum. The source supplies the out-of-equilibrium driving and is the only edge with no precursor vertex.
10. **Sink (fixed-sink edge).** $(\sigma, n, p, \mathbf{c}) \rightarrow \emptyset$ at a fixed sink (free surface, grain boundary acting as a perfect sink, or a dislocation). The sink rate is set by sink strengths and absorbs both polarities.

Admissibility rules. A vertex $(\sigma, n, p, \mathbf{c})$ is admissible if and only if (i) the polarity is consistent with the population (e.g., a “vacancy-loop” population p admits $\sigma = -1$ only); (ii) the composition lies in the host-specific admissible region $\mathcal{C}_{\sigma, n, p} \subseteq \mathcal{C}$ (e.g., a helium–vacancy cluster respects the maximum self-trapping ratio); and (iii) the size is at or above the minimum size for the population (monomers always exist in the bulk population; loops exist only for $n \geq n_{\text{loop}}^{\min}$). An edge is admissible if and only if its endpoints are admissible and its stoichiometric tuple respects the host conservation laws.

2.3. Hierarchical Subgraph Decomposition

The vertex set decomposes hierarchically as

$$\mathcal{V} = \mathcal{V}_- \cup \mathcal{V}_+ = \left(\bigcup_{p \in \mathcal{P}_-} \mathcal{V}_-^p \right) \cup \left(\bigcup_{p \in \mathcal{P}_+} \mathcal{V}_+^p \right). \quad (8)$$

The two outermost blocks $\mathcal{V}_-$ and $\mathcal{V}_+$ are the *polarity layers*: vacancy-class clusters and SIA-class clusters. Within each polarity layer, the vertex set is partitioned into *population layers* $\mathcal{V}_\sigma^p$, one for each population label $p \in \mathcal{P}_\sigma$. The composition coordinate $\mathbf{c}$ indexes vertices within a population layer; it is not a layer of its own because solute trapping and detrapping are intra-population reactions.

The edge set decomposes accordingly:

$$\mathcal{E} = \underbrace{\mathcal{E}_{--} \cup \mathcal{E}_{++}}_{\text{intra-polarity edges}} \cup \underbrace{\mathcal{E}_{-+} \cup \mathcal{E}_{+-}}_{\text{cross-polarity edges (recombination)}}. \quad (9)$$

Within each intra-polarity block,

$$\mathcal{E}_{\sigma\sigma} = \left(\bigcup_{p \in \mathcal{P}_\sigma} \mathcal{E}_{\sigma\sigma}^{p,p} \right) \cup \left(\bigcup_{\substack{p, p' \in \mathcal{P}_\sigma \\ p \neq p'}} \mathcal{E}_{\sigma\sigma}^{p,p'} \right), \quad (10)$$

where the first term collects intra-population edges (growth, shrinkage, dissociation, solute trapping, intra-population coalescence) and the second collects inter-population edges (segregation, sweeping, capture by moving sinks,

ejection by re-resolution). Cross-polarity edges $\mathcal{E}_{\pm\mp}$ are, by their nature, inter-vertex but always intra-population at the surviving vertex (recombination at a vacancy cluster does not relocate the vacancy cluster). The decomposition (8)–(10) exposes the structural sparsity of the master equations: most edges in $\mathcal{E}$ reside within an intra-population block $\mathcal{E}_{\sigma\sigma}^{p,p}$; the strong intra-population coupling and the comparatively weak inter-population and cross-polarity coupling are the sources of the “bordered-banded” Jacobian structure.

2.4. Specialization to Host Materials

The same abstract graph $\mathcal{G}$ specializes to a target host by declaring its admissible vertex and edge sets. Two contrasting cases illustrate the modularity.

EUROFER-97 (bcc Fe, fusion-spectrum first wall). The vertex set is

$$\mathcal{V}_-^{\text{EUR}} = \{ (-1, n, \text{bulk}, (c_{\text{He}}, c_{\text{H}})) : n \geq 1, \text{ adm.} \}, \quad (11)$$

$$\mathcal{V}_+^{\text{EUR}} = \{ (+1, n, \text{bulk-}\langle 111 \rangle, \mathbf{0}) \} \cup \{ (+1, n, \text{bulk-}\langle 100 \rangle, \mathbf{0}) \}. \quad (12)$$

The $\mathcal{V}_-$ layer is composition-rich (helium, hydrogen) but single-population (only the bulk vacancy cluster is tracked explicitly; grain-boundary segregation is folded into a sink strength). The $\mathcal{V}_+$ layer is composition-trivial but multi-population, distinguishing $\langle 111 \rangle$ and $\langle 100 \rangle$ SIA loops.

Zirconium alloys (hcp Zr, fission cladding). The vertex set is

$$\mathcal{V}_-^{\text{Zr}} = \{ (-1, n, \text{prism. } \langle a \rangle, \mathbf{0}) \} \cup \{ (-1, n, \text{basal} \langle c \rangle, \mathbf{0}) \}, \quad (13)$$

$$\mathcal{V}_+^{\text{Zr}} = \{ (+1, n, \text{prism. } \langle a \rangle, \mathbf{0}) \}. \quad (14)$$

The $\mathcal{V}_-$ layer carries two populations because $\langle c \rangle$ -loops are vacancy clusters distinct from prismatic $\langle a \rangle$ vacancy loops, with very different binding energies and inter-population kinetics. Inter-population edges $\mathcal{E}_{--}^{\text{void}, \langle c \rangle}$ encode the nucleation-driven population conversion that controls breakaway growth in fuel cladding. Helium and hydrogen are absent from the active state space.

3. Generalized Defect-Cluster Population Dynamics

3.1. Abstract Model Structure

Let the primitive irradiation-produced species be $\mathcal{E} = \{I, V, \mathbf{g}\}$, where I denotes a self-interstitial atom, V denotes a vacancy, and $\mathbf{g}$ denotes an

ordered list of gas species. The gas list is written as $\mathbf{g} = (g_1, g_2, \dots, g_{N_g})$, where N_g is the number of gas species included in the model. Gas atoms in the matrix are represented by reservoir concentrations

$$c_\mu^g(t), \quad \mu = 1, \dots, N_g, \quad (15)$$

where μ labels the gas species g_μ . These variables represent monomeric gas atoms in the matrix reservoir. They are coupled to vacancy clusters through trapping, emission, re-solution, and gas release from eliminated cavities. The model does not include reactions of the type: $g_\mu + g_\nu \longrightarrow$ gas cluster, so gas-only clusters are excluded.

The formulation distinguishes between a *species class* and a *defect population*. A species class identifies the broad physical type of defect cluster. A defect population identifies a subcollection of clusters within a species class that share the same mobility law, orientation state, microstructural environment, or sink coupling. The interstitial-cluster species class is denoted by $\mathcal{C}_I$. Its defect populations are denoted by

$$\mathcal{P}_I^\alpha \subset \mathcal{C}_I, \quad \alpha = 1, \dots, N_P^I. \quad (16)$$

The label α is a population label, not a cluster-size index. For example, different interstitial-cluster populations may represent mobile SIA clusters, sessile SIA loops, loops aligned with a principal stress direction, or loops not aligned with that direction.

The vacancy-cluster species class is denoted by $\mathcal{C}_V$. Its defect populations are denoted by

$$\mathcal{P}_V^\beta \subset \mathcal{C}_V, \quad \beta = 1, \dots, N_P^V. \quad (17)$$

The label β is also a population label. For example, different vacancy-cluster populations may represent bulk cavities, dislocation-attached cavities, grain-boundary cavities, or surface-attached cavities.

An interstitial-cluster state is written as

$$I_{n,\alpha}, \quad n = 1, \dots, N_I, \quad \alpha = 1, \dots, N_P^I, \quad (18)$$

where n is the number of self-interstitial atoms and α identifies the interstitial-cluster population. Its concentration is $c_{n,\alpha}^I(t)$. A vacancy-cluster, or cavity, state is written as

$$V_{m,\ell,\beta}, \quad m = 1, \dots, N_V, \quad \ell \in \mathcal{G}_m, \quad \beta = 1, \dots, N_P^V. \quad (19)$$

Here m is the number of vacancies in the cavity, β identifies the vacancy-cluster population, and

$$\boldsymbol{\ell} = (\ell_1, \ell_2, \dots, \ell_{N_g}) \in \mathbb{Z}_{\geq 0}^{N_g} \quad (20)$$

is the gas inventory trapped inside the cavity. The component ℓ_μ is the number of atoms of gas species g_μ trapped inside the cavity. The set $\mathcal{G}_m$ denotes the admissible gas occupancies for a cavity containing m vacancies. Its concentration is $c_{m,\boldsymbol{\ell},\beta}^V(t)$. The notation $V_{m,\boldsymbol{\ell},\beta}$ denotes a vacancy-cluster or cavity state. Conventional chemical notation may be recovered as a special case. For example, if $\mathbf{g} = (\text{He})$, then $V_{m,(\ell),\beta}$ denotes $V_m\text{He}_\ell$ in population β . If $\mathbf{g} = (\text{He}, \text{T})$, then $V_{m,(\ell,p),\beta}$ denotes $V_m\text{He}_\ell\text{T}_p$ in population β . A gas-free cavity is obtained by setting $\boldsymbol{\ell} = \mathbf{0}$; $V_{m,\mathbf{0},\beta}$ denotes a pure vacancy cluster V_m in population β .

The full deterministic state vector is therefore

$$\mathbf{c}(t) = \left[\{c_{n,\alpha}^I\}_{n,\alpha}, \{c_{m,\boldsymbol{\ell},\beta}^V\}_{m,\boldsymbol{\ell},\beta}, \{c_\mu^g\}_{\mu=1}^{N_g} \right]^\top. \quad (21)$$

The first block contains interstitial-cluster populations, the second block contains vacancy-cluster/cavity populations, and the third block contains matrix gas-reservoir variables.

The generic signed-collision rule is

$$I_{n,\alpha} + V_{m,\boldsymbol{\ell},\beta} \longrightarrow \begin{cases} I_{n-m,\alpha'} + \mathcal{R}_\ell, & n > m, \\ \mathcal{R}_\ell, & n = m, \\ V_{m-n,\boldsymbol{\ell},\beta'}, & m > n. \end{cases} \quad (22)$$

Here α' and β' denote the product populations, determined by the specific reaction rule. In the simplest single-population model, $\alpha' = \alpha$ and $\beta' = \beta$ may be suppressed. The operator $\mathcal{R}_\ell$ denotes the release of the gas inventory from an eliminated cavity to the matrix gas reservoir. In the default reservoir model, $\mathcal{R}_\ell = \sum_{\mu=1}^{N_g} \ell_\mu g_\mu$. If $m > n$, the cavity survives and retains its gas inventory ℓ . If $n \geq m$, the supporting vacancy cluster is eliminated and its gas content is returned to the gas reservoir.

3.2. The Reaction-Network Graph (RAG)

The state space defined above contains all cluster states that may be represented by the model, but not every formal combination of states corre-

sponds to a physically possible reaction. A reaction may be forbidden by crystallography, defect orientation, Burgers-vector compatibility, elastic interaction symmetry, migration dimensionality, topology, or energetics. Therefore, the construction of the master equation requires two distinct steps: (1) define the state space of possible defect clusters; (2) define the admissible reaction graph on that state space.

The master equation is assembled from elementary reactions. Each reaction is assigned an index r . The index r labels one of the ten specific physical processes, which are described in Section 2.2. Let X_a denote a generic state in the model. The state index a may refer to an interstitial cluster, a vacancy cluster, or a gas-reservoir species:

$$X_a \in \{I_{n,\alpha}, V_{m,\ell,\beta}, g_\mu\}. \quad (23)$$

Thus a is not a physical size by itself. It is a compact label for one state variable in the ODE system.

For a given reaction r , let ν_{ar}^- be the number of objects of state X_a consumed by the reaction, and let ν_{ar}^+ be the number of objects of state X_a produced by the reaction. The net change of state X_a caused by one occurrence of reaction r is $S_{ar} = \nu_{ar}^+ - \nu_{ar}^-$. The quantity S_{ar} is the stoichiometric coefficient of state a in reaction r . A general reaction can therefore be written compactly as

$$\sum_a \nu_{ar}^- X_a \longrightarrow \sum_a \nu_{ar}^+ X_a. \quad (24)$$

For example, the interstitial-cluster coalescence reaction

$$I_{n,\alpha} + I_{n',\alpha'} \longrightarrow I_{n+n',\alpha''} \quad (25)$$

has $S_{I_{n,\alpha},r} = -1$, $S_{I_{n',\alpha'},r} = -1$, $S_{I_{n+n',\alpha''},r} = +1$, with all other entries of S_{ar} equal to zero. Similarly, gas trapping by a cavity,

$$g_\mu + V_{m,\ell,\beta} \longrightarrow V_{m,\ell+\mathbf{e}_\mu,\beta'}, \quad (26)$$

has $S_{g_\mu,r} = -1$, $S_{V_{m,\ell,\beta},r} = -1$, $S_{V_{m,\ell+\mathbf{e}_\mu,\beta'},r} = +1$. Here $\mathbf{e}_\mu$ is the unit vector in gas-species space.

3.3. The Master Equation as a “Graph Walker”

For a state X_a on a graph, the concentration rate of change is assembled by a graph traversal procedure, which, for convenience, we will refer to as a

reaction-graph *walker*. The walker visits each admissible reaction edge $e \in \mathcal{E}$, reads its stoichiometric update vector $\boldsymbol{\nu}_e$ and rate law k_e , and deposits the associated gain–loss contributions into the master equation. The concentration of state X_a is denoted by $c_a(t)$. In terms of the explicit variables,

$$c_a \in \{c_{n,\alpha}^I, c_{m,\ell,\beta}^V, c_\mu^g\}. \quad (27)$$

The mean-field master equation is

$$\boxed{\frac{dc_a}{dt} = G_a + \sum_r S_{ar} J_r(\mathbf{c}) - D_a c_a.} \quad (28)$$

Here G_a is the direct production rate of state a from collision cascades, $J_r(\mathbf{c})$ is the flux of reaction r , and $D_a c_a$ is a first-order loss to unresolved fixed sinks. Equation (28) states that the concentration of state a changes by summing the contributions of all reactions that produce or consume that state.

The reaction flux J_r depends on the reaction type. For a unary transition, $X_b \longrightarrow \text{products}$, $J_r(\mathbf{c}) = \Gamma_r c_b$, where Γ_r is a first-order rate coefficient. For a binary collision, $X_b + X_c \longrightarrow \text{products}$, $J_r(\mathbf{c}) = K_{bc}^{(r)} c_b c_c$, where $K_{bc}^{(r)}$ is the appropriate collision or capture rate. For a direct source process, $\emptyset \longrightarrow X_a$, $J_r = G_r$. The generalized model is built from the following admissible process classes.

Irradiation production. Displacement cascades and gas production processes introduce source terms $G_a = G_a^{\text{FP}} + G_a^{\text{cascade}} + G_a^{\text{gas}} + G_a^{\text{trans}}$, where the terms denote Frenkel-pair production, clustered cascade production, direct gas production, and transmutation or implantation sources.

Interstitial-interstitial coalescence. Interstitial clusters may coalesce according to

$$I_{n,\alpha} + I_{n',\alpha'} \longrightarrow I_{n+n',\alpha''), \quad (29)$$

with flux

$$J_{n\alpha,n'\alpha'}^{II} = K_{n\alpha,n'\alpha'}^{II} c_{n,\alpha}^I c_{n',\alpha'}^I. \quad (30)$$

The product population α'' is specified by the physical population rule. For a single interstitial population, the population indices are suppressed.

Vacancy-vacancy coalescence. Vacancy clusters may coalesce according to

$$V_{m,\ell,\beta} + V_{m',\ell',\beta'} \longrightarrow V_{m+m',\ell+\ell',\beta''}. \quad (31)$$

The vacancy numbers add, and the gas occupancy vectors add component-wise. The product population β'' is determined by the population rule for the coalescence event.

Interstitial-vacancy collisions. Cross-species collisions between interstitial and vacancy clusters obey Eq. (22). These events conserve signed lattice-defect content and conserve each gas species after the matrix reservoir is included.

Gas trapping by cavities. A gas monomer of species μ may be trapped by a vacancy cluster:

$$g_\mu + V_{m,\ell,\beta} \longrightarrow V_{m,\ell+\mathbf{e}_\mu,\beta'}. \quad (32)$$

The corresponding flux is

$$J_{m\ell\beta,\mu}^{\text{trap}} = K_{m\ell\beta,\mu}^{\text{trap}} c_\mu^g c_{m,\ell,\beta}^V. \quad (33)$$

Gas emission and radiation re-solution. Gas may be thermally emitted or athermally re-solved from a cavity:

$$V_{m,\ell,\beta} \longrightarrow V_{m,\ell-\mathbf{e}_\mu,\beta'} + g_\mu, \quad \ell_\mu \geq 1. \quad (34)$$

The corresponding flux may include both thermal and athermal contributions:

$$J_{m\ell\beta,\mu}^{\text{rel}} = [\varepsilon_\mu(m, \ell, \beta) + \Gamma_\mu^{\text{res}}(m, \ell, \beta)] c_{m,\ell,\beta}^V. \quad (35)$$

Trap mutation. An over-pressurized gas-bearing cavity may emit an SIA while increasing its vacancy content:

$$V_{m,\ell,\beta} \longrightarrow V_{m+1,\ell,\beta'} + I_{1,\alpha'}. \quad (36)$$

This reaction conserves the signed lattice-defect content because $-m = -(m+1) + 1$.

Thermal vacancy and interstitial emission. Vacancy emission from a cavity is represented by

$$V_{m,\ell,\beta} \longrightarrow V_{m-1,\ell,\beta'} + V_{1,0,\beta''}, \quad m \geq 2. \quad (37)$$

SIA emission from an interstitial cluster is represented by

$$I_{n,\alpha} \longrightarrow I_{n-1,\alpha'} + I_{1,\alpha''}, \quad n \geq 2. \quad (38)$$

Both processes conserve signed lattice-defect content.

Population transfer. A cluster may move between defect populations without changing its size or gas inventory. For interstitial clusters, $I_{n,\alpha} \longrightarrow I_{n,\alpha'}$, and for vacancy clusters, $V_{m,\ell,\beta} \longrightarrow V_{m,\ell,\beta'}$. These transitions may represent changes in mobility, stress orientation, attachment to a dislocation, capture by a grain boundary, or transfer between bulk and microstructural reservoirs. They conserve signed defect content and gas inventory.

Loss to unresolved sinks. Unresolved sinks are represented by first-order losses: $X_a \longrightarrow \emptyset$, $J_a^{\text{sink}} = D_a c_a$. When sinks are unbiased, their contributions cancel in the signed-defect balance. When sinks are biased, their net contribution must be accumulated as a separate sink-flux correction.

The elementary process classes are summarized in Figure 1 as a reaction network graph. The figure layers four conceptual elements introduced earlier: the species classes $\mathcal{C}_V$ and $\mathcal{C}_I$, the defect populations $\mathcal{P}_V^\beta$ and $\mathcal{P}_I^\alpha$ within each class, the point-defect monomers V_1 and I_1 together with the gas reservoir, and the linear sinks (dislocation network, grain boundary, precipitate). Edges encode the admissible reactions: production from a displacement cascade or transmutation source, intra-population growth and shrinkage, inter-population transfer, cross-class recombination, gas trapping and release between cavities and the matrix gas reservoir, and absorption at linear sinks.

The master equation may be decomposed into production, binary collision, unary transition, gas exchange, population-transfer, and sink terms:

$$\dot{c}_a = G_a + \mathcal{Q}_a(\mathbf{c}, \mathbf{c}) + \mathcal{L}_{ab} c_b + \mathcal{H}_a(\mathbf{c}, \mathbf{c}^g) + \mathcal{P}_{ab} c_b - D_a c_a. \quad (39)$$

Here $\mathcal{Q}_a$ collects binary cluster-cluster collisions, $\mathcal{L}_{ab}$ collects unary emission, trap mutation, and re-resolution operators, $\mathcal{H}_a$ collects gas-reservoir exchange terms, and $\mathcal{P}_{ab}$ collects population-transfer processes. This decomposition is only organizational; all of these terms are generated by the single stoichiometric form in Eq. (28).

The binary collision operator may be written as

$$\mathcal{Q}_a(\mathbf{c}, \mathbf{c}) = \frac{1}{2} C_{abc}^+ K_{bc} c_b c_c - C_{abc}^- K_{bc} c_b c_c, \quad (40)$$

where repeated state indices are summed. The tensor C_{abc}^+ identifies binary reactions in which states b and c produce state a , while C_{abc}^- identifies binary reactions in which state a is consumed by collision with another state. The factor $1/2$ avoids double counting of symmetric collisions.

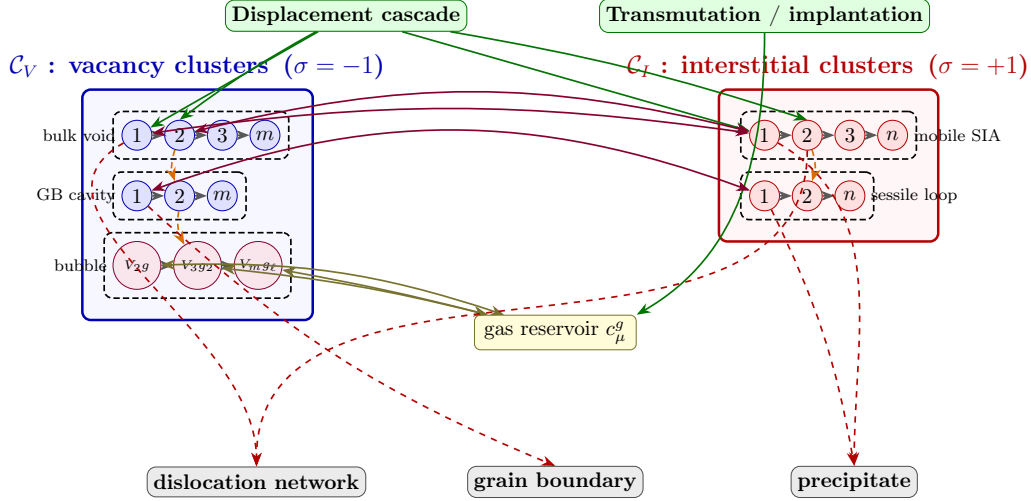


Figure 1: Reaction Network Graph (RAG) for the generalized cluster-dynamics formulation. The two outer enclosures are the species classes $\mathcal{C}_V$ (vacancy clusters, polarity $\sigma = -1$, blue) and $\mathcal{C}_I$ (interstitial clusters, polarity $\sigma = +1$, red). Each class is partitioned into defect populations (dashed boxes): {bulk void, GB cavity, bubble} for the vacancy class and {mobile SIA, sessile loop} for the interstitial class. Within each population, vertices are clusters indexed by size; the leftmost vertex in the bulk and mobile-SIA populations is the corresponding point-defect monomer (V_1 and I_1). The matrix gas reservoir c_μ^g (yellow) sits between the two classes; the displacement cascade and transmutation sources (green, top) inject clusters into the network. Edges carry six classes of admissible reactions. The graph is the host-independent template.

4. The Ferritic–Martensitic Steels System

The abstract framework of Sections 2.1–3 specializes in the reduced-activation ferritic–martensitic steel EUROFER-97 in three explicit steps. Step 1 (Section 4.1) declares the host-specific reaction admissibility graph $\mathcal{G}^{\text{EUR}}$: admissible vertex set, active populations, and gas list. Step 2 (Section 4.3) enumerates the eight admissible edge classes P1–P8 and their closed-form rate kernels. Step 3 (Section 4.4) applies the abstract graph-walker accumulator to each population.

4.1. State-Space Declaration

Gas list and composition vector. For EUROFER-97 under neutron irradiation, the only resolved trapped solute is helium:

$$N_g = 1, \quad g_1 = \text{He}, \quad \ell = (\ell), \quad (41)$$

so the gas reservoir $c_1^g(t) \equiv c_h(t)$ and the composition vector reduce to the scalar helium occupancy $\ell \in \mathbb{Z}_{\geq 0}$.

Active populations. EUROFER-97 carries one population per polarity:

$$\mathcal{P}_-^{\text{EUR}} = \{ \text{bulk} \}, \quad N_P^V = 1, \quad (42)$$

$$\mathcal{P}_+^{\text{EUR}} = \{ \text{bulk} \}, \quad N_P^I = 1. \quad (43)$$

The single SIA population covers both 3D-mobile clusters ($n < 4$) and $\frac{1}{2}\langle 111 \rangle$ prismatic loops ($n \geq 4$); the distinction is encoded in size-dependent rate kernels, not a separate population label. Grain boundaries, the dislocation network, and MX/M₂₃C₆ precipitates are treated as fixed unresolved sinks. With one population per polarity the population labels are suppressed:

$$c_{n,\alpha}^I(t) \rightarrow c_n(t), \quad c_{m,(\ell),\beta}^V(t) \rightarrow c_{m,\ell}(t), \quad c_1^g(t) \rightarrow c_h(t). \quad (44)$$

Admissible vertex set.

$$\mathcal{V}_-^{\text{EUR}} = \{ (-1, m, \text{bulk}, (\ell)) : 1 \leq m \leq N_V, 0 \leq \ell \leq \ell_{\max}(m) \}, \quad (45)$$

$$\mathcal{V}_+^{\text{EUR}} = \{ (+1, n, \text{bulk}, \mathbf{0}) : 1 \leq n \leq N_I \}, \quad (46)$$

$$\mathcal{V}_0^{\text{EUR}} = \{ g_1 = \text{He} \}, \quad (47)$$

comprising size-resolved SIA clusters, size-resolved voids ($\ell = 0$) and helium bubbles ($\ell \geq 1$), and the single matrix He reservoir. The maximum helium content obeys the saturation rule

$$\ell_{\max}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor, \quad \alpha_{\text{He}} \in [1.5, 2.0]. \quad (48)$$

Three primary populations. Equations (45)–(47) yield three populations evolved by the master equations: (1) **SIA clusters** $c_n(t)$, $n = 1, \dots, N_I$: loops with $\frac{1}{2}\langle 111 \rangle$ Burgers vector for $n \geq 4$, and 3D-mobile clusters for $n = 1, 2, 3$, carrying no helium; (2) **vacancy clusters** / **cavities** $c_{m,\ell}(t)$, $m = 1, \dots, N_V$, $\ell = 0, \dots, \ell_{\max}(m)$: spherical cavities, pure voids when $\ell = 0$ and helium bubbles when $\ell \geq 1$; (3) **free helium** $c_h(t)$, the single gas-reservoir variable. The signed lattice-defect contents are

$$q(I_n) = +n, \quad q(V_m \text{He}_\ell) = -m, \quad q(\text{He}) = 0, \quad (49)$$

as required by the conservation law of Section 4.5.

4.2. Reaction Rates for Ferritic-Martensitic Steels

This section derives the rate constants for all reactions between defect clusters, expressed in atomic-fraction units (s^{-1}) and in terms of the reaction frequencies $\omega_\alpha = D_\alpha/a^2$ and the geometric prefactor $\xi(m) = (3m/8\pi)^{1/3}$ introduced in the diffusion section. The cluster notation (m, ℓ) and concentration units (at/at) follow the conventions established in the diffusion treatment of the cluster-dynamics formulation.

For a mobile point defect α absorbed by an immobile spherical cluster of m defects, the Smoluchowski volumetric rate constant [42] is $\mathcal{K}^{\text{vol}} = 4\pi r_m (D_A + D_B)$. Converting to atomic-fraction units ($\mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega$) and substituting $D = \omega a^2$, $r_m = \xi(m) a$, and $\Omega = a^3/2$:

$$\mathcal{K}_{\alpha,m}^{3D} = 8\pi \xi(m) \omega_\alpha \quad [\text{s}^{-1}]. \quad (50)$$

When both reactants are mobile, the rate constant includes both frequencies:

$$\mathcal{K}_{AB}^{3D} = 8\pi \xi_{\text{rec}} (\omega_A + \omega_B), \quad (51)$$

with $\xi_{\text{rec}} = \sqrt{3}/2 \approx 0.866$ for nearest-neighbor capture in bcc.

Vacancy–SIA recombination ($m = n = 1$, both mobile):

$$\mathcal{K}_{iv} = 4\sqrt{3}\pi (\omega_i + \omega_v) \approx 21.8\omega_i, \quad (52)$$

since $\omega_i \gg \omega_v$ at all relevant temperatures.

Point defect absorption by a spherical cluster (void or bubble) of m vacancies:

$$\mathcal{K}_{\alpha,m}^{\text{void}} = A_{\text{sph}} m^{1/3} \omega_\alpha, \quad A_{\text{sph}} = (48\pi^2)^{1/3} \approx 7.818. \quad (53)$$

This is the recommended code-ready form: a single precomputed constant multiplied by $m^{1/3}$ and the appropriate reaction frequency.

Point defect absorption by a dislocation loop of n SIAs requires species-specific rate constants because only SIAs experience a non-trivial elastic bias ($Z_i^{\text{loop}} > 1$), while vacancies are unbiased ($Z_v^{\text{loop}} = 1$):

$$\mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i, \quad (54)$$

$$\mathcal{K}_{v,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} \omega_v, \quad (55)$$

with $A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \approx 10.78$ and $Z_i^{\text{loop}} \approx 1.02$ – 1.10 (Table 3). SIA absorption grows the loop by one; vacancy absorption shrinks it by one.

Point defect absorption rate by the dislocation network (first-order sink): $-Z_\alpha \rho_d a^2 \omega_\alpha c_\alpha$, where ρ_d (m^{-2}) is the dislocation line density, and no n_{at} conversion is needed because this term is first-order in c_α .

SIA clusters of size $n \geq 4$ migrate by 1D glide along $\langle 111 \rangle$ with the size-dependent SIA-cluster reaction frequency:

$$\omega_n^{1D} = \frac{3}{2} \cdot \frac{\nu_0}{n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad s \approx 0.5-1.0, \quad E_m^{1D} \lesssim 0.05 \text{ eV}. \quad (56)$$

The effective 3D reaction frequency after Burgers vector rotation is:

$$\omega_n^{3D, \text{eff}} = \frac{\omega_n^{1D} L}{3\sqrt{3}a}. \quad (57)$$

where L is the mean free path between rotations[43]. The rate constant for an n -SIA cluster reacting with an immobile void of m vacancies is then:

$$\mathcal{K}_{n,m}^{\text{mixed}} = A_{\text{sph}} m^{1/3} \omega_n^{3D, \text{eff}} = \frac{A_{\text{sph}}}{3\sqrt{3}} m^{1/3} \hat{L} \omega_n^{1D} \approx 1.505 m^{1/3} \hat{L} \omega_n^{1D}, \quad (58)$$

where $\hat{L} = L/a$ is the dimensionless mean free path.

For a one-dimensionally migrating cluster reacting with a random distribution of spherical sinks, the appropriate rate coefficient is not generally the ordinary three-dimensional Smoluchowski coefficient. In the pure correlated 1D limit, the sink strength is quadratic in the projected sink density. For sink family m ,

$$\left. \frac{dc_n}{dt} \right|_m = -D_n^{\text{eff}} k_{n,m}^2 c_n, \quad (59)$$

where D_n^{eff} is the effective diffusivity used in the rate equation and $k_{n,m}^2$ is the sink strength for absorption of the mobile cluster by sink family m . In the pure 1D limit [44],

$$k_{(1),m}^2 = 6\kappa_m \kappa, \quad \kappa_m = \sigma_m N_m, \quad \kappa = \sum_j \kappa_j, \quad (60)$$

where N_m is the number density of sinks of type m , σ_m is their geometrical absorption cross section, and κ^{-1} is the mean 1D diffusion range in the sink field. For a monodisperse population of spherical sinks with radius r_m ,

$$\sigma_m = \pi r_m^2, \quad k_{(1),m}^2 = 6\pi^2 r_m^4 N_m^2. \quad (61)$$

Thus, since $r_m \propto m^{1/3}$, the pure 1D sink strength scales as $m^{4/3}N_m^2$, in contrast to the ordinary 3D sink strength,

$$k_{(3),m}^2 = 4\pi r_m N_m, \quad (62)$$

which scales as $m^{1/3}N_m$.

For finite direction-change length, the transition from correlated 1D kinetics to 3D kinetics can be represented by the Trinkaus–Heinisch–Barashev–Golubov–Singh master-curve interpolation [44]. Let ℓ_n denote the mean straight-line migration length of cluster n between changes of its 1D migration direction,

$$\ell_n^2 = D_n^{\text{lo}} \tau_{\text{rot},n}, \quad (63)$$

where D_n^{lo} is the longitudinal 1D diffusivity and $\tau_{\text{rot},n}$ is the mean time between direction changes. The effective sink strength is then written as

$$k_{n,m}^2 = y_{n,m} k_{(1),m}^2, \quad (64)$$

with

$$y_{n,m} = \frac{1}{2} \left[1 + (1 + 4x_{n,m}^2)^{1/2} \right], \quad (65)$$

and

$$x_{n,m} = \left[\frac{\ell_n^2 k_{(1),m}^2}{12} + \frac{k_{(1),m}^4}{k_{(3),m}^4} \right]^{-1/2}. \quad (66)$$

This form recovers the correct limiting behaviors. When $\ell_n \gg \kappa^{-1}$, direction changes are rare, $x_{n,m} \rightarrow 0$, $y_{n,m} \rightarrow 1$, and therefore

$$k_{n,m}^2 \rightarrow k_{(1),m}^2. \quad (67)$$

When direction changes are very frequent, $\ell_n \ll r_m$, the migration becomes effectively three-dimensional and the interpolation gives

$$k_{n,m}^2 \rightarrow k_{(3),m}^2. \quad (68)$$

The loss term may therefore be written compactly as

$$\left. \frac{dc_n}{dt} \right|_m = -D_n^{\text{eff}} y_{n,m} k_{(1),m}^2 c_n. \quad (69)$$

The pure 1D limit is quadratic in sink density and scales as r_m^4 , whereas the 3D limit is first order in sink density and scales as r_m . This difference makes

the absorption of one-dimensionally migrating clusters especially sensitive to sink size, sink density, and the direction-change mean free path.

Thermal emission of species α from a cluster of size n is related to the capture rate by detailed balance. The daughter cluster has size $(n - 1)$, so the capture rate that enters the detailed-balance relation uses $(n - 1)$:

$$\alpha_\alpha(n) = A_{\text{sph}} (n - 1)^{1/3} \omega_\alpha \exp\left(-\frac{E_b^\alpha(n)}{k_B T}\right) \quad [\text{s}^{-1}], \quad (70)$$

where $E_b^\alpha(n)$ is the binding energy of the last defect of type α to the cluster (from detailed-balance dissociation). The emission enters the master equation as a first-order process:

$$\left. \frac{dc_n}{dt} \right|_{\text{emission}} = +\alpha_\alpha(n) c_n - \alpha_\alpha(n + 1) c_{n+1}. \quad (71)$$

For a dislocation loop emitting an SIA, replace $A_{\text{sph}} (n - 1)^{1/3}$ with $A_{\text{loop}} (n - 1)^{1/2}$:

$$\alpha_i^{\text{loop}}(n) = A_{\text{loop}} (n - 1)^{1/2} \omega_i \exp\left(-\frac{E_b^i(n)}{k_B T}\right). \quad (72)$$

For $\text{He}_\ell \text{V}_m$ bubbles, three emission channels exist (detailed-balance for the He–bubble system):

Vacancy emission:

$$\varepsilon_v(m, \ell) = A_{\text{sph}} (m - 1)^{1/3} \omega_v \exp\left(-\frac{E_b^v(m, \ell)}{k_B T}\right). \quad (73)$$

Helium emission:

$$\varepsilon_h(m, \ell) = A_{\text{sph}} m^{1/3} \omega_h \exp\left(-\frac{E_B^{\text{He}}(m, \ell)}{k_B T}\right). \quad (74)$$

He emission uses $m^{1/3}$ (not $(m - 1)^{1/3}$) because removal of a He atom does not change the vacancy count; the capture radius for detailed balance is that of the daughter $\text{He}_{\ell-1} \text{V}_m$, which retains the same m . The binding energy E_B^{He} is from the He–vacancy detailed-balance relation summarised in this section.

Trap mutation (Frenkel pair emission):

$$\Gamma_{\text{TM}}(m, \ell) = \nu_0 \exp\left(-\frac{E_{\text{TM}}(m, \ell)}{k_B T}\right), \quad (75)$$

producing $\text{He}_\ell \text{V}_{m+1} + \text{SIA}$. Table 3 collects the rate-constant prefactors, sink parameters, and state-space controls used in the reaction-rate expressions (Section 4.2).

4.3. Admissible Edges and Rate Kernels

Process classes P1–P8. The admissible edges of $\mathcal{G}^{\text{EUR}}$ ($\chi_{ab \rightarrow c} = 1$, & $\chi_{a \rightarrow b} = 1$) partition into eight process classes, each mapped to an abstract edge class in Section 2.2. There is only one SIA loop population here; hence, the intra-population transfer edge is suppressed and the defect production edge is treated as a source term.

Process	Abstract edge class	Rate kernel
P1: V–SIA recombination	cross-polarity (recombination)	$\mathcal{K}_{iv}$
P2: cavity absorption of mono- α	growth / shrinkage	$\mathcal{K}_{\alpha,m}^{\text{cav}}$
P3: loop absorption of mono- α	growth / shrinkage	$\mathcal{K}_{\alpha,n}^{\text{loop}}$
P4: absorption at fixed sink	sink edge	$\mathcal{D}_{\alpha}$
P5: thermal monomer emission	dissociation	ε_{α}
P6: 1D-gliding SIA cluster $\rightarrow$ cavity	cross-polarity	$\mathcal{K}_{n,m}^{\text{eff}}$
P7: trap mutation	gas-pressure-driven	$\Gamma_{\text{TM}}(m, \ell)$
P8: radiation re-solution	solute detrapping	$\Gamma_{\text{res}}(m, \ell)$

Mobility cutoffs. Mobile clusters are bounded by size limits from Table 3: SIA clusters are mobile (3D or 1D) for $1 \leq n \leq i_{\text{mobile}}$; larger clusters are sessile dislocation loops acting only as fixed sinks. Vacancy clusters are 3D mobile for $1 \leq m \leq v_{\text{mobile}}$; larger cavities grow or shrink only through single-monomer absorption or emission. Under the mono-defect simplification ($i_{\text{mobile}} = v_{\text{mobile}} = 1$), the coalescence rate constants reduce to

$$\mathcal{K}_{n,1}^{ii}|_{n'=1} = \mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}, \quad (76)$$

$$\mathcal{K}_{m,1}^{vv}|_{m'=1} = \mathcal{K}_{v,m}^{\text{cav}} = A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}}, \quad (77)$$

$$\mathcal{K}_{m=1,n=1}^{vi} = \mathcal{K}_{iv} = 4\sqrt{3} \pi (\omega_i^{\text{eff}} + \omega_v^{\text{eff}}). \quad (78)$$

Reaction Processes (general coalescence formulation). The most complete description allows any two mobile clusters to react. The three intra-polarity and cross-polarity channels are:

SIA–SIA coalescence ($I_n + I_{n'} \rightarrow I_{n+n'}$, $1 \leq n' \leq n \leq i_{\text{mobile}}$):

$$\mathcal{K}_{n,n'}^{ii} = 8\pi (\xi_n + \xi_{n'}) (\omega_n^{\text{eff}} + \omega_{n'}^{\text{eff}}), \quad (79)$$

with $\xi_n = (3n/8\pi)^{1/3}$. *Vacancy-vacancy coalescence* ($V_m + V_{m'} \rightarrow V_{m+m'}$, $m, m' \leq v_{\text{mobile}}$):

$$\mathcal{K}_{m,m'}^{vv} = 8\pi (\xi_m + \xi_{m'}) (\omega_m^{\text{eff}} + \omega_{m'}^{\text{eff}}). \quad (80)$$

V-I cluster annihilation ($V_m + I_n \rightarrow V_{m-n}, I_{n-m}$, or $\emptyset$):

$$\mathcal{K}_{m,n}^{vi} = 8\pi (\xi_m + \xi_n) (\omega_m^{\text{eff}} + \omega_n^{\text{eff}}), \quad m \leq v_{\text{mobile}}, n \leq i_{\text{mobile}}. \quad (81)$$

He-He coalescence is excluded because the interstitial He-He binding energy in bcc Fe is negligible ($\lesssim 0.05$ eV).

4.4. RAG Assembly of the Master Equations

The abstract master equation 28 is assembled algorithmically by looping over every admissible edge $r \in \mathcal{E}^{\text{EUR}}$, computing the flux J_r , and accumulating the signed contribution $S_{ar}J_r$ into each affected population a . Table 1 lists every edge class P1-P8, its reaction, its flux, and its non-zero stoichiometric contributions to the three EUROFER-97 population equations $\dot{c}_n$, $\dot{c}_{m,\ell}$, and $\dot{c}_h$. Where a process connects adjacent nodes on a ladder (e.g. cavity growth/shrinkage), the gain term at a and the loss term at a appear on separate rows; the net effect is the tridiagonal ladder operator visible in the boxed equations. Figure 2 shows the full EUROFER-97 reaction network as a directed graph, with the eight process classes color-coded on the arrows and the graph-walker formula annotated in the upper center.

Reading Table 1 column by column yields the three ODEs for the concentrations of SIA clusters, vacancy-helium clusters, and free helium: $\dot{c}_n$ receives contributions from P1 (loss at $n = 1$), P2i/P6 (loss to cavities), P3 (i-i and v-i ladders), P4 (sink loss), P5i (emission ladder), and P7 (trap-mutation SIA source at $n = 1$). $\dot{c}_{m,\ell}$ receives contributions from P2v (vacancy absorption ladder in m), P2i/P6 (SIA-induced shrinkage), P2h and P5h (He ladders in ℓ), P3 (vacancy side), P4 (mobile-cavity sink loss), P5v (vacancy emission), P7 (trap-mutation ladder), and P8 (re-resolution ladder). $\dot{c}_h$ receives contributions from P2h (absorption loss), P4 (sink loss), P5h (emission gain), and P8 (re-resolution gain). Every term in the master equations below has a one-to-one correspondence to a row and column in Table 1.

Define the total cavity concentration at vacancy size m and the combined fixed-sink rate:

$$\bar{c}_m(t) \equiv \sum_{\ell=0}^{\ell_{\text{max}}(m)} c_{m,\ell}(t), \quad \mathcal{D}_\alpha = \mathcal{D}_\alpha^d + \mathcal{D}_\alpha^{gb} + \mathcal{D}_\alpha^p. \quad (82)$$

Table 1: Graph-walker contribution table for EUROFER-97 (mono-defect, $i_{\text{mobile}} = v_{\text{mobile}} = 1$). Each row shows one edge class, its reaction, the corresponding flux J_r , and the sign of the contribution $S_{ar}J_r$ to each population. The symbol $\Delta_m^\pm$ denotes a ladder step: $\Delta_m^+ f_m \equiv f_{m-1} - f_m$ (gain from smaller cavity minus loss at m) and $\Delta_m^- f_m \equiv f_{m+1} - f_m$ (gain from larger cavity minus loss at m). Sums over repeated size indices are implied

P	Reaction	Flux J_r	$\Delta \dot{c}_n$	$\Delta \dot{c}_{m,\ell}$	$\Delta \dot{c}_h$
P1	$I_1 + V_1 \rightarrow 0$	$\mathcal{K}_{iv} c_1 c_{1,0}$	$-J$ at $n=1$	$-J$ at $(1, 0)$	0
P2v (growth)	$V_1 + V_m \rightarrow V_{m+1}$	$\mathcal{K}_{v,m}^{\text{cav}} c_{1,0} c_{m,\ell}$	0	$\Delta_m^+ J$ ladder	0
P2i (shrinkage)	$I_n + V_m \rightarrow V_{m-n}$	$\mathcal{K}_{n,m}^{(\text{cav})} c_n c_{m,\ell}$	$-\sum_m J$ at n	SIA-shrinkage ladder	0
P2h (He trapping)	$\text{He} + V_{m,\ell} \rightarrow V_{m,\ell+1}$	$\mathcal{K}_{h,m}^{\text{cav}} c_h c_{m,\ell}$	0	He-occ. ladder in ℓ	$-\sum_{m,\ell < \ell_{\text{max}}} J$
P3 (loop growth)	$I_1 + I_n \rightarrow I_{n+1}$	$\mathcal{K}_{i,n}^{\text{loop}} c_1 c_n$	i-i ladder in n	0	0
P3 (loop shrink)	$V_1 + I_n \rightarrow I_{n-1}$	$\mathcal{K}_{v,n}^{\text{loop}} c_{1,0} c_n$	v-i ladder in n	0	0
P4 (sink loss)	$X_\alpha \rightarrow \emptyset$	$\mathcal{D}_\alpha c_\alpha$	$-\mathcal{D}_i(n) c_n$	$-\mathcal{D}_v c_{m,\ell} \delta_{m \leq v_{\text{mob}}}$	$-\mathcal{D}_h c_h$
P5v (emission)	$V_m \rightarrow V_{m-1} + V_1$	$\varepsilon_v(m, \ell) c_{m,\ell}$	0	ε_v ladder in m	0
P5i (emission)	$I_n \rightarrow I_{n-1} + I_1$	$\varepsilon_i(n) c_n$	ε_i ladder in n	0	0
P5h (emission)	$V_{m,\ell} \rightarrow V_{m,\ell-1} + \text{He}$	$\varepsilon_h(m, \ell) c_{m,\ell}$	0	He-occ. ladder in ℓ	$+\sum_{m,\ell \geq 1} \varepsilon_h c_{m,\ell}$
P6 (1D glide)	$I_n^{1D} + V_m \rightarrow V_{m-n}$	$\mathcal{K}_{n,m}^{\text{eff}} c_n c_{m,\ell}$	$-\sum_m J$ at $n \geq 4$	1D-shrinkage ladder	0
P7 (trap mutation)	$\text{He}_\ell V_m \rightarrow \text{He}_\ell V_{m+1} + I_1$	$\Gamma_{\text{TM}}(m, \ell) c_{m,\ell}$	$+\sum_{m,\ell} \Gamma_{\text{TM}} c_{m,\ell}$ at $n=1$	TM ladder in m	0
P8 (re-solution)	$\text{He}_\ell V_m \rightarrow \text{He}_{\ell-1} V_m + \text{He}$	$b_0 \dot{\phi} \ell c_{m,\ell}$	0	re-sol. ladder in ℓ	$+\sum_{m,\ell \geq 1} b_0 \dot{\phi} \ell c_{m,\ell}$
G_a	cascade sources	$G_n, G_{m,\ell}, G_{\text{He}}$	$+G_n$	$+G_{m,\ell}$	$+G_{\text{He}}$

The equations below are written in the general (coalescence) form; the mono-defect simplification is recovered by setting $i_{\text{mobile}} = v_{\text{mobile}} = 1$ (Eqs. 76–78).

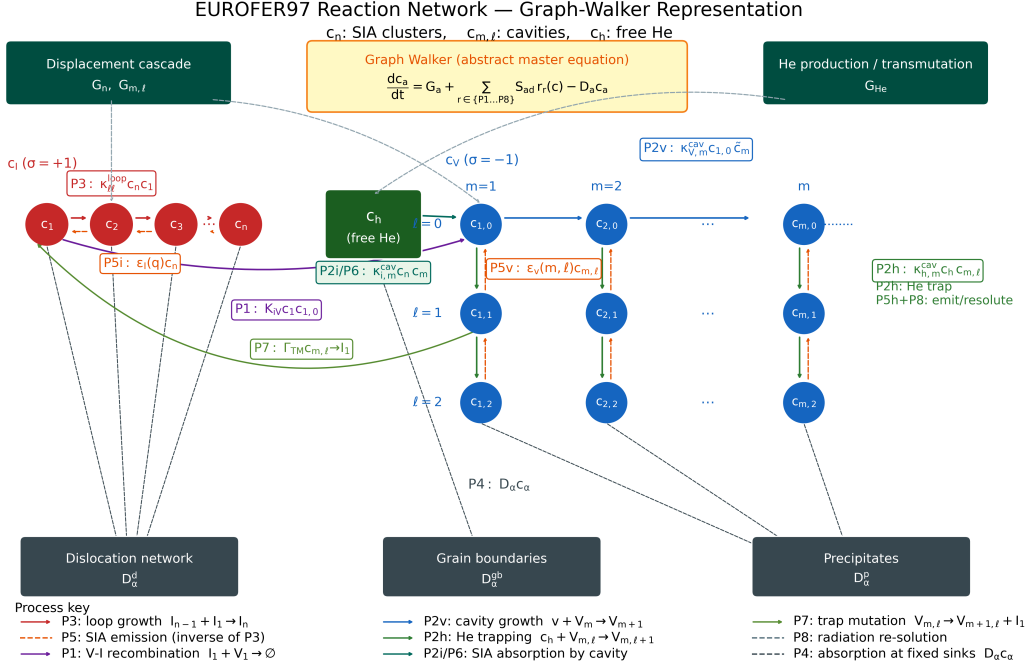


Figure 2: Reaction network for the EUROFER-97 system. Nodes represent the three evolving populations: SIA clusters c_n (left column, **I**, blue), free helium c_h (center, **He**, orange), and vacancy/He cavities $c_{m,l}$ (right grid, V_m - V_m He $_l$, green/purple). The cascade source (top) and fixed sinks (bottom) are shown as boxes. Directed arrows are color-coded by process class P1–P8 and labeled with the corresponding rate kernel. The annotated formula in the upper center is the graph-walker $\dot{c}_a = G_a + \sum_r S_{ar} J_r - D_a c_a$ (Eq. 28); each labeled arrow in the figure corresponds to one term $S_{ar} J_r$ in that sum. Walking the graph along all eight process classes recovers the master equations (83)–(86)

Defect Class 1: SIA clusters c_n .

$$\begin{aligned}
\frac{dc_n}{dt} = & \underbrace{G_n}_{\text{production}} + \underbrace{\frac{1}{2} \sum_{n'=1}^{\min(n-1, i_{\text{mobile}})} \mathcal{K}_{n', n-n'}^{ii} c_{n'} c_{n-n'}}_{\text{SIA-SIA coalescence gain } (I_{n'} + I_{n-n'} \rightarrow I_n)} \\
& - \underbrace{c_n \sum_{n'=1}^{i_{\text{mobile}}} \mathcal{K}_{n, n'}^{ii} c_{n'}}_{\text{SIA-SIA coalescence loss } (I_n + I_{n'} \rightarrow I_{n+n'})} + \underbrace{\varepsilon_i(n+1) c_{n+1} - \varepsilon_i(n) c_n}_{\text{thermal emission (SIA from loops, P5)} \delta_{n \geq 2}} \\
& + \underbrace{\sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m', n+m'}^{vi} c_{m', 0} c_{n+m'}}_{\text{V-I annihilation gain } (V_{m'} + I_{n+m'} \rightarrow I_n)} - \underbrace{\sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m', n}^{vi} c_{m', 0} c_n}_{\text{V-I annihilation loss } (V_{m'} + I_n \rightarrow I_{n-m'})} \\
& - \underbrace{\sum_{m=1}^{N_V} \mathcal{K}_{n, m}^{(\text{cav})} c_n \bar{c}_m}_{\text{SIA cluster } \rightarrow \text{cavity absorption (P2/P6)}} - \underbrace{\mathcal{D}_i(n) c_n}_{\text{fixed-sink loss (mobile clusters only, P4)}} \\
& + \underbrace{\delta_{n=1} \sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} \Gamma_{\text{TM}}(m, \ell) c_{m, \ell}}_{\text{trap-mutation SIA source at } n=1 \text{ (P7)}}
\end{aligned} \tag{83}$$

where the cavity rate constant $\mathcal{K}_{n, m}^{(\text{cav})}$ is

$$\mathcal{K}_{n, m}^{(\text{cav})} = \begin{cases} A_{\text{sph}} m^{1/3} \omega_i^{\text{eff}}, & 1 \leq n \leq 3 \text{ (3D, P2)}, \\ \frac{A_{\text{sph}} m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}, & 4 \leq n \leq i_{\text{mobile}} \text{ (1D/3D, P6)}, \\ 0, & n > i_{\text{mobile}} \text{ (sessile)} \end{cases} \tag{84}$$

Defect Class 2: Vacancy clusters $c_{m, \ell}$. Voids ($\ell = 0$) and bubbles ($\ell \geq 1$) share a unified (m, ℓ) grid. Spherical cavities are unbiased ($Z_i^{\text{cav}} = Z_v^{\text{cav}} = 1$). The v-v coalescence terms are active only for $m \leq v_{\text{mobile}}$; SIA-induced shrinkage spans all mobile SIA sizes through $\mathcal{K}_{n, m}^{(\text{cav})}$.

$$\begin{aligned}
\frac{dc_{m,\ell}}{dt} = & \underbrace{G_{m,\ell}}_{\text{production}} + \underbrace{\frac{1}{2} \sum_{m'=1}^{\min(m-1, v_{\text{mobile}})} \mathcal{K}_{m',m-m'}^{vv} c_{m',0} c_{m-m',\ell} \delta_{m \leq 2 v_{\text{mobile}}}}_{\text{V-V coalescence gain } (V_{m'} + V_{m-m'} \rightarrow V_m)} \\
& - \underbrace{c_{m,\ell} \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m,m'}^{vv} c_{m',0}}_{\text{V-V coalescence loss } (V_m + V_{m'} \rightarrow V_{m+m'})} + \underbrace{\varepsilon_v(m+1, \ell) c_{m+1,\ell} - \varepsilon_v(m, \ell) c_{m,\ell}}_{\text{thermal emission (vacancy from cavities, P5)}} \\
& + \underbrace{\sum_{n=1}^{\min(i_{\text{mobile}}, N_V - m)} \mathcal{K}_{n,m+n}^{(\text{cav})} c_n c_{m+n,\ell}}_{\text{SIA-induced shrinkage gain } (I_n + V_{m+n} \rightarrow V_m)} - \underbrace{\sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n c_{m,\ell}}_{\text{SIA-induced shrinkage loss } (I_n + V_m \rightarrow V_{m-n})} \\
& + \underbrace{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell-1} \delta_{\ell \geq 1} - A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell} \delta_{\ell < \ell_{\text{max}}}}_{\text{He absorption by cavities (P2, He-side)}} \\
& + \underbrace{\varepsilon_h(m, \ell+1) c_{m,\ell+1} - \varepsilon_h(m, \ell) c_{m,\ell} \delta_{\ell \geq 1}}_{\text{He emission from cavities (P5, He-side)}} + \underbrace{\Gamma_{\text{TM}}(m-1, \ell) c_{m-1,\ell} - \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}}_{\text{trap mutation (P7)}} \\
& + \underbrace{b_0 \dot{\phi}(\ell+1) c_{m,\ell+1} - b_0 \dot{\phi} \ell c_{m,\ell}}_{\text{radiation re-solution (P8)}} - \underbrace{\mathcal{D}_v c_{m,\ell} \delta_{m \leq v_{\text{mobile}}}}_{\text{fixed-sink loss (mobile cavities only, P4)}}
\end{aligned} \tag{85}$$

Free helium c_h . He-He coalescence is excluded, and helium interacts with the vacancy-cluster grid only through mono-He trapping and emission.

$$\begin{aligned}
\frac{dc_h}{dt} = & \underbrace{G_{\text{He}}}_{\text{production}} - \underbrace{\sum_{m=1}^{N_V} A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \sum_{\ell=0}^{\ell_{\text{max}}(m)-1} c_{m,\ell}}_{\text{He absorption by cavities (P2)}} \\
& - \underbrace{\mathcal{D}_h c_h}_{\text{fixed-sink loss (P4)}} + \underbrace{\sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} \varepsilon_h(m, \ell) c_{m,\ell}}_{\text{He emission (P5)}} + \underbrace{\sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} b_0 \dot{\phi} \ell c_{m,\ell}}_{\text{radiation re-solution (P8)}}
\end{aligned} \tag{86}$$

4.5. Mass Conservation Laws & the Swelling Identity

The master equations (83)–(86) satisfy two exact conservation identities. Their derivation rests on three facts that are built into the RADCLUSTER code:

- (i) cascade production deposits SIA and vacancy atoms in a balanced Frenkel-pair ratio, $\sum_{n=1}^{N_I} n P_n^{(i)} = \sum_{m=1}^{N_V} m P_m^{(v)} = \eta G$
- (ii) every mutual reaction $I_n + V_m$ destroys $\min(n, m)$ SIA atoms *and* $\min(n, m)$ vacancy atoms; the residual fragment (V_{m-n} if $m > n$, I_{n-m} if $n > m$) survives in its corresponding population. The two destruction sums therefore cancel identically in $d(S_I - S)/dt$;
- (iii) only mobile clusters with $n \leq n_{\text{max}}^i$ (SIAs) or $m \leq m_{\text{max}}^v$ (vacancies) reach fixed sinks. The corresponding rate-equation losses are summed over the full mobile range, not the monomer alone.

Cumulative loss to fixed sinks. The total fixed-sink rate for species $\alpha \in \{i, v, h\}$ collects all three sink channels, $\mathcal{D}_\alpha(n) = \mathcal{D}_\alpha^d(n) + \mathcal{D}_\alpha^{gb}(n) + \mathcal{D}_\alpha^p(n)$, and vanishes outside the mobile range. Define the cumulative atomic content

delivered to fixed sinks:

$$\Delta J_i^{\text{fix}}(t) \equiv \int_0^t \sum_{n=1}^{n_{\text{max}}^i} n \mathcal{D}_i(n) c_n(t') dt' \geq 0, \quad (87)$$

$$\Delta J_v^{\text{fix}}(t) \equiv \int_0^t \sum_{m=1}^{m_{\text{max}}^v} m \mathcal{D}_v \bar{c}_m(t') dt' \geq 0, \quad (88)$$

$$\Delta J_h^{\text{fix}}(t) \equiv \int_0^t \left[\mathcal{D}_h c_h + \mathcal{D}_v \sum_{m=1}^{m_{\text{max}}^v} \sum_{\ell} \ell c_{m,\ell} \right] dt' \geq 0. \quad (89)$$

These three integrals are appended to the state vector and evolved as auxiliary ODEs, so that they share the integrator's local error control with the physical concentrations.

Frenkel-pair balance. With total inventories

$$S_I(t) \equiv \sum_{n=1}^{N_I} n c_n(t), \quad S(t) \equiv \sum_{m=1}^{N_V} m \bar{c}_m(t), \quad (90)$$

facts (i)–(iii) combine into the exact Frenkel-pair conservation law

$$S(t) - S_I(t) - \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t) = S(0) - S_I(0) = \text{const.} \quad (91)$$

For a defect-free initial condition, this is the *swelling identity*

$$S(t) = S_I(t) + \Delta J_i^{\text{fix}}(t) - \Delta J_v^{\text{fix}}(t), \quad (92)$$

i.e. the total vacancy content of all cavities equals the SIA inventory in loops plus the *net* atomic flux delivered to fixed sinks. Crucially, the SIA flux to fixed sinks [ΔJ_i^{fix} , Eq. (87)] sums the contributions of *all* mobile SIA cluster sizes $n = 1, \dots, n_{\text{max}}^i$, and the vacancy flux [ΔJ_v^{fix} , Eq. (88)] sums the contributions of *all* mobile vacancy cluster sizes $m = 1, \dots, m_{\text{max}}^v$, not just the monomer terms. Differentiating gives the rate form

$$\frac{dS}{dt} = \frac{dS_I}{dt} + \sum_{n=1}^{n_{\text{max}}^i} n \mathcal{D}_i(n) c_n - \sum_{m=1}^{m_{\text{max}}^v} m \mathcal{D}_v \bar{c}_m. \quad (93)$$

Once loops saturate ($dS_I/dt \approx 0$), swelling grows at the rate at which the biased dislocation channel ($Z_i^d \neq Z_v^d$) drains the surplus SIA inventory faster than the vacancy inventory; the unbiased grain-boundary and precipitate channels contribute equally to both sums and cancel in $dS/dt - dS_I/dt$ only insofar as the mobile SIA and vacancy populations are equal.

Helium conservation. The total helium inventory is the sum of free and trapped helium,

$$S_{\text{He}}(t) \equiv c_h(t) + \sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\max}(m)} \ell c_{m,\ell}(t). \quad (94)$$

Trap mutation (P7), radiation re-resolution (P8), and thermal He emission (P5h) only redistribute helium between c_h and the trapped pool, so their contributions to dS_{He}/dt cancel pairwise. The only net helium sinks are free helium captured at fixed sinks ($\mathcal{D}_h c_h$) and helium carried to fixed sinks by mobile cavities ($m \leq m_{\max}^v$); immobile bubbles retain their helium. Hence

$$\frac{dS_{\text{He}}}{dt} = G_{\text{He}} - \mathcal{D}_h c_h - \mathcal{D}_v \sum_{m=1}^{m_{\max}^v} \sum_{\ell} \ell c_{m,\ell}, \quad (95)$$

i.e. $dS_{\text{He}}/dt = G_{\text{He}} - d\Delta J_h^{\text{fix}}/dt$. In particular, S_{He} is *not* conserved against $\int G_{\text{He}} dt'$ alone — the fixed-sink loss must be tracked explicitly. Both loss terms are typically small (mobile clusters carry little He) but they are non-zero and must appear in the diagnostic below for it to converge to zero.

Numerical accuracy diagnostics. The swelling identity (92) and the helium balance (95) provide two inexpensive a-posteriori checks, neither of which requires post-processing because the auxiliary integrals (87)–(89) are already state variables:

$$\delta_{\text{FP}}(t) = \frac{|S(t) - S_I(t) - \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t)|}{S(t) + S_I(t) + \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t)}, \quad (96)$$

$$\delta_{\text{He}}(t) = \frac{\left| S_{\text{He}}(t) + \Delta J_h^{\text{fix}}(t) - \int_0^t G_{\text{He}} dt' - S_{\text{He}}(0) \right|}{S_{\text{He}}(t) + \Delta J_h^{\text{fix}}(t) + \int_0^t G_{\text{He}} dt'}. \quad (97)$$

Both should remain below $\sim 10^{-6}$ in a correctly implemented code; values above 10^{-3} almost always signal a coding error, missing or incorrect reactions, or an overly loose integrator tolerance. Despite the importance of these conservation identities, they have seldom been discussed or tested in the cluster dynamics literature.

5. State Space Reduction

5.1. Phase Space Reduction Strategy

Transmission electron microscopy of ferritic-martensitic steels irradiated to 50–100 dpa in fission and fusion-relevant spectra establishes the physical scale of the cluster populations. Dislocation loops ($\frac{1}{2}\langle 111 \rangle$ SIA clusters) grow to mean diameters of 10–30 nm at temperatures of 350–500 °C; the largest loops resolved at high dose approach 50 nm. Converting to SIA content, a 30 nm loop contains $n \approx 1.3 \times 10^4$, and a 50 nm loop $n \approx 3.7 \times 10^4$. The SIA size ladder therefore runs to $N_I \sim 10^3$ – 10^4 . Helium bubbles, continuously nucleated by the 10–15 appm He/dpa transmutation source of a fusion neutron spectrum, stabilize at diameters of 1–3 nm under typical operating conditions (350–450 °C); a 1 nm bubble corresponds to $m \approx 45$ vacancies, and a 3 nm bubble to $m \approx 1200$. Near the swelling peak (480–550 °C), a subset of cavities grow to 5–10 nm ($m \approx 5 \times 10^3$ – 4×10^4), so the vacancy ladder also spans $N_V \sim 10^3$ – 10^4 .

The full state vector has three blocks:

$$n_{\text{eq}}^{\text{full}} = \underbrace{N_I}_{\text{SIA ladder}} + \underbrace{n_{VH}^{\text{full}}}_{\text{vacancy-He grid}} + \underbrace{1}_{c_h}. \quad (98)$$

The SIA block is a simple 1-D ladder of N_I equations. The vacancy–He block is two-dimensional: every vacancy size $m = 1, \dots, N_V$ carries a helium sub-ladder of depth $\ell_{\text{max}}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor$ (Eq. 48), so the total vacancy–He grid size grows super-linearly with N_V :

$$n_{VH}^{\text{full}} = \sum_{m=1}^{N_V} [\ell_{\text{max}}(m) + 1] \approx N_V \left(1 + \frac{3}{5} \alpha_{\text{He}} N_V^{2/3} \right), \quad (99)$$

reaching 10^4 – 10^5 for $N_V = 10^3$ – 10^4 . Combined with the SIA class, the unreduced system involves $n_{\text{eq}}^{\text{full}} \sim 10^4$ – 2×10^5 coupled ODEs.

The right-hand side of this system is severely stiff; an implicit BDF method is mandatory, requiring the solution of a $n_{\text{eq}} \times n_{\text{eq}}$ Newton system at each time step. A dense LU factorization costs $O(n_{\text{eq}}^3)$ per step. At $n_{\text{eq}} = 10^5$ that is $\sim 10^{15}$ floating-point operations per Newton solve — roughly 10^{10} seconds on a modern processor, rendering a single irradiation run impossible in practice. Even with the sparse, bordered-banded Jacobian structure exploited in RADCLUSTER, each RHS evaluation sweeps over all

(m, ℓ) pairs, so the cost scales as $O(n_{VH}^{\text{full}})$ per step. For a parameter study over temperature, dose rate, or He/dpa ratio, even a factor-of-ten reduction in system size yields a hundred-fold speedup: reduction is not an optional refinement but a prerequisite for practical simulation.

We exploit the physical and mathematical structure of the master equations to derive reduced systems that are *exact in an appropriate limit* or *systematically improvable*. Physics and numerical phase space reduction methods are currently available in RADCLUSTER, as follows:

Physics:

1. **Fast-He mean-field approximation (fusion conditions)** Helium in iron has a migration energy of only $E_m^h \approx 0.06$ eV [45], giving a He jump rate that exceeds the vacancy jump rate by a factor of $\sim 10^6$ at 400 °C. Under fusion-relevant He generation (10-15 appm/dpa), He redistributes within and between cavities on a timescale far shorter than the growth timescale of the cavities themselves. The full 2-D distribution $c_{m,\ell}$ therefore collapses: at each vacancy size m , the He content equilibrates to a mean value $\bar{\ell}(m, t)$. The 2-D grid n_{VH}^{full} is replaced by two 1-D arrays $(\bar{c}_m, Q_m)$, reducing the vacancy-He block to $2N_V$ equations (Section 5.2).
2. **Decoupled He inventory (fission conditions)** Fission reactors produce only 0.1–1 appm/dpa of helium; most cavities are pure voids with $\bar{\ell} \ll m$, and He acts as a weak perturbation on void kinetics rather than a nucleation driver. In this regime, the per-cluster He content $\bar{\ell}(m)$ need not be tracked individually: the total trapped helium $Q_{\text{tot}}(t)$ evolves as a single scalar ODE, and the distribution among sizes is recovered algebraically from the capture-rate weighting (Section 5.3). The vacancy–He block shrinks to $N_V + 1$ equations.

Size-bin moment reduction (numerical, both conditions):

Even after the He approximations above, the SIA and vacancy size ladders still run to $N_I, N_V \sim 10^3$ – 10^4 . The bin-moment method partitions each ladder into $N_B \sim 20$ – 60 logarithmically spaced bins and replaces the per-size ODEs with equations for the zeroth and first moments within each bin. This reduces the $O(N)$ per-species ladder to $O(N_B) = O(\log N)$ equations with

no assumption on the He kinetics and is the primary strategy for parametric studies and upscaled simulations (Section 5.4).

The remainder of this section derives the equations for these three systematic methods of reduction. Throughout this section, we use the notation and rate constants established in Sections 4.2 and 4: $\omega_\alpha^{\text{eff}}$ are the effective reaction frequencies (Eq. 125); $A_{\text{sph}} = (48\pi^2)^{1/3}$; $\bar{c}_m = \sum_\ell c_{m,\ell}$ is the total cavity concentration; and $\mathcal{K}_{n,m}^{(\text{cav})}$ is the SIA-cluster absorption rate (Eq. 84). The vacancy net-flux from size m to $m+1$ is defined as:

$$J_{m \rightarrow m+1} \equiv A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0} \bar{c}_m - \varepsilon_v(m+1, \bar{\ell}(m+1)) \bar{c}_{m+1}, \quad (100)$$

with forward (capture) and backward (emission) components $J_{m \rightarrow m+1}^+$ and $J_{m \rightarrow m+1}^-$ respectively, so that $J_{m \rightarrow m+1} = J_{m \rightarrow m+1}^+ - J_{m+1 \rightarrow m}^-$.

5.2. Fast He Equilibration — Fusion Conditions

At temperatures where vacancies are mobile ($T \gtrsim 200^\circ\text{C}$), the He reaction frequency greatly exceeds the vacancy reaction frequency $\omega_h^{\text{eff}} \gg \omega_v^{\text{eff}}$. Helium equilibrates on each cavity faster than the cavity grows or shrinks, so the full two-dimensional (m, ℓ) distribution can be collapsed into two scalar quantities per vacancy size: the total cavity concentration $\bar{c}_m$ and the mean He loading

$$\bar{\ell}(m, t) \equiv \frac{\sum_\ell \ell c_{m,\ell}}{\bar{c}_m}. \quad (101)$$

Summing Eq. (85) over all ℓ at fixed m :

$$\frac{d\bar{c}_m}{dt} = J_{m-1 \rightarrow m} - J_{m \rightarrow m+1} - \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m + \Gamma_{\text{TM}}(m-1, \bar{\ell}) \bar{c}_{m-1} - \Gamma_{\text{TM}}(m, \bar{\ell}) \bar{c}_m, \quad (102)$$

where the vacancy net flux $J_{m \rightarrow m+1}$ is given by Eq. (100). The SIA-shrinkage sum $\sum_n \mathcal{K}_{n,m}^{(\text{cav})} c_n$ spans all mobile SIA clusters $n = 1, \dots, i_{\text{mobile}}$. The vacancy capture rate $A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}}$ is independent of ℓ , so the ℓ -summation is exact for the capture terms. The emission rate ε_v depends on ℓ through the He gas-pressure correction to $E_b^v(m, \ell)$ and is approximated by evaluation at the mean loading $\bar{\ell}(m)$.

Define the He inventory at vacancy size m as $N_{\text{He}}(m) \equiv \bar{\ell}(m) \bar{c}_m$. Multiplying Eq. (85) by ℓ , summing over ℓ , and collecting terms:

$$\begin{aligned}
\frac{dN_{\text{He}}(m)}{dt} = & A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \bar{c}_m - \varepsilon_h(m, \bar{\ell}) \bar{c}_m - b_0 \dot{\phi} N_{\text{He}}(m) \\
& + \bar{\ell}(m-1) J_{m-1 \rightarrow m}^+ - \bar{\ell}(m) J_{m \rightarrow m+1}^+ \\
& + \bar{\ell}(m+1) J_{m+1 \rightarrow m}^- - \bar{\ell}(m) J_{m \rightarrow m-1}^- \\
& - \bar{\ell}(m) \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m - \mathcal{D}_v N_{\text{He}}(m) \delta_{m \leq v_{\text{mobile}}}.
\end{aligned} \tag{103}$$

The five groups of terms are: (i) He captured from the free pool minus thermal He emission (P5) minus radiation re-solution (P8); (ii–iii) He is advected in and out as clusters grow or shrink along the m -axis, where J^+ is the forward (capture) component and J^- is the backward (emission) component of the vacancy flux; (iv) He is carried away when SIA clusters annihilate vacancies; and (v) He is lost when mobile vacancy clusters ($m \leq v_{\text{mobile}}$) are absorbed at fixed sinks.

Applying the quotient rule $d\bar{\ell}/dt = (\dot{N}_{\text{He}} - \bar{\ell} \dot{\bar{c}}_m)/\bar{c}_m$ and substituting Eq. (102):

$$\begin{aligned}
\frac{d\bar{\ell}(m)}{dt} = & A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h - \varepsilon_h(m, \bar{\ell}(m)) - b_0 \dot{\phi} \bar{\ell}(m) \\
& + \frac{J_{m-1 \rightarrow m}^+}{\bar{c}_m} [\bar{\ell}(m-1) - \bar{\ell}(m)] + \frac{J_{m+1 \rightarrow m}^-}{\bar{c}_m} [\bar{\ell}(m+1) - \bar{\ell}(m)].
\end{aligned} \tag{104}$$

Line 1 contains the local He kinetics: trapping from the free pool (proportional to c_h), thermal emission ε_h (P5), and athermal re-solution (P8), which drives $\bar{\ell}$ toward zero. Line 2 contains advection corrections: cavities flowing into size class m carry their own He loading, which may differ from $\bar{\ell}(m)$; the difference $[\bar{\ell}(m \pm 1) - \bar{\ell}(m)]$ acts as a drift source.

5.3. Decoupled He Inventory — Fission Conditions

Under fission irradiation, the He generation rate is low (~ 0.3 – 1 appm/dpa vs. ~ 10 – 15 appm/dpa for fusion), so most cavities carry $\ell \ll m$ and the He/V ratio $\bar{\ell}/m \ll 1$. Helium acts as a weak perturbation on the void kinetics rather than controlling nucleation. Rather than tracking $\bar{\ell}(m)$ as a separate variable at each size, the total trapped He inventory is followed as a single

scalar and distributed algebraically across cavity sizes. The $\bar{c}_m$ equation is identical to Case 1, Eq. (102), with the vacancy emission rate evaluated using a He-corrected binding energy. In the low-loading limit, the He gas pressure provides an effective binding energy.

Define $\ell_{\text{tot}} \equiv \sum_{m=1}^M \ell(m) \bar{c}_m$. Summing the He content over all cavities and using Eq. (86) with thermal emission dropped:

$$\frac{d\ell_{\text{tot}}}{dt} = A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m=1}^M m^{1/3} \bar{c}_m - b_0 \dot{\phi} \ell_{\text{tot}} - \mathcal{D}_v \sum_{m=1}^{v_{\text{mobile}}} \ell(m) \bar{c}_m. \quad (105)$$

The first term is the total He trapping rate by all cavities (the sum is the weighted capture cross-section $\sum_m m^{1/3} \bar{c}_m$), the second is the athermal re-solution loss (P8), and the third is He lost when mobile vacancy clusters ($m \leq v_{\text{mobile}}$) are absorbed at fixed sinks. Helium is then distributed across cavity sizes in proportion to the capture rate $A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$, as that determines the steady-state He loading when trapping and re-solution balance.

$$\ell(m) = \ell_{\text{tot}} \frac{m^{1/3} \bar{c}_m}{\sum_{m'=1}^M m'^{1/3} \bar{c}_{m'}}. \quad (106)$$

This algebraic relation is recomputed at each time step. The $m^{1/3}$ weighting arises from the spherical-cavity capture radius $\xi(m) \propto m^{1/3}$, consistent with $\mathcal{K}_{h,m} = A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$.

The free-He equation retains all terms from Eq. (86), including the fixed-sink loss $\mathcal{D}_h c_h$, but with thermal He emission dropped:

$$\frac{dc_h}{dt} = G_{\text{He}} - A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m=1}^M m^{1/3} \bar{c}_m + b_0 \dot{\phi} \ell_{\text{tot}} - \mathcal{D}_h c_h. \quad (107)$$

The trapping and re-solution terms are exactly the negatives of those in Eq. (105), so the He conservation law $d(c_h + \ell_{\text{tot}})/dt = G_{\text{He}} - \mathcal{D}_h c_h$ holds.

5.4. State-Space Reduction via Size-Bin Moments

Coarse-graining (grouping) of the cluster size-distribution function (SDF) has a long lineage. Various grouping methods have been advanced during the past decades [2, 11, 35, 46–49]. The earliest scheme [46] partitions size space

into geometrically widening groups, but it introduces spurious broadening of the SDF and a drifting mean size. The moment-conserving scheme of [50] removed the mass-leakage problem by describing each group with two variables—a zeroth moment (number density) and a slope—so that both the group number and mass are conserved; it is robust but, as originally formulated, is restricted to monomer fluxes and cannot natively treat cluster–cluster coalescence. A complementary deterministic route couples a discrete treatment at small sizes to a Fokker–Planck drift–diffusion continuum at large sizes through a matched transition region [51], which is accurate for specific cases but whose general applicability hinges on the matching procedure. [52] unified and generalized these ideas in a basis-function framework that subsumes two families: (i) *interpolation* schemes (linear, secant, Lagrange, cubic-spline) in which the SDF is a piecewise polynomial and the grouped reaction coefficients are derived analytically *once* before integration under the assumption of constant rate constants within a group; and (ii) a *generalized moment* scheme extending [50] to arbitrary reaction networks—cluster–cluster coalescence, mobile and one-dimensionally migrating clusters, and genuinely multi-component phase spaces such as He bubbles. A philosophically orthogonal reduction is the hybrid deterministic/stochastic coupling of [53], which keeps deterministic cluster dynamics in the dense small-size region and switches to stochastic sampling for the sparse large-size tail, trading statistical variance for resolution rather than coarse-graining the size axis at all.

The present work (RADCLUSTER; [41]) occupies a distinct point in this design space: it is a hybrid discrete plus bin-moment "reconstruct–project" scheme in which small sizes are resolved discretely, and larger sizes are placed in logarithmic bins carrying one, two, or three moments under a piecewise-constant, linear dual-basis, or lognormal closure. Unlike [50, 52], it precomputes no analytic grouped coefficients; instead, each right-hand-side evaluation reconstructs the full per-size SDF, applies the *exact* size-dependent master equation—including the one-/three-dimensional mobile-cluster transitions and cluster–cluster reactions—and reprojects onto the bin moments, so the grouped system is exactly consistent with the discrete model by construction. Helium is not treated as a gridded phase-space axis (as in the [52] bubble problem) but through an orthogonal mean-loading state-space reduction, and conservation is *monitored* via dedicated Frenkel-pair and helium accounting integrals (δ_{FP} , δ_{He}) rather than enforced structurally. The trade-off is that only the state-vector dimension—not the per-step arithmetic—is

reduced, an expense offset by sparse-linear-algebra and preconditioner engineering; in return, the three-moment lognormal closure supplies within-bin curvature ($\mu_k^{(2)}$) that has no analog in either the [50] or [52] schemes.

The full per size master equations (Sec. 4) are replaced by a much smaller ODE system in which the smallest cluster sizes are still resolved with one ODE per size (*discrete range*), while the larger sizes are aggregated into logarithmically spaced bins (*moment range*). Three intra-bin closures are supported in the RADCLUSTER code: piecewise-constant ($P = 1$ moment per bin), hat-function / linear dual basis ($P = 2$), and log-normal ($P = 3$). Both the SIA and the vacancy populations are reduced in the same way; the He state is attached to the vacancy bins (Case 1) or carried as a scalar total (Case 2). The total ODE count drops from $O(I + V)$ to

$$N_{\text{eq}} = \underbrace{i_{\text{discrete}} + P I_{\text{bin}}}_{\text{SIA block}} + \underbrace{v_{\text{discrete}} + P V_{\text{bin}}}_{\text{vacancy block}} + n_{\text{He}} + 5, \quad (108)$$

where the trailing 5 counts the cumulative-loss accounting integrals (Sec. 4.5). Typically, 20–60 bins suffice for $I, V = 10^4$ – 10^5 with $r \in [1.5, 2]$.

The SIA size axis $\{1, 2, \dots, I\}$ is partitioned into a *discrete* prefix $\{1, \dots, i_{\text{discrete}}\}$ that is tracked one ODE per size, followed by I_{bin} geometrically spaced bins that cover $\{i_{\text{discrete}}+1, \dots, I\}$. The discrete cutoff i_{discrete} is a numerical parameter chosen independently of the physical mobility cutoff n_{max}^i ; in particular, i_{discrete} may exceed n_{max}^i if extra resolution at small immobile loops is desired. The bin edges $\{s_k\}_{k=0}^{I_{\text{bin}}}$ satisfy

$$s_0 = i_{\text{discrete}} + 1, \quad s_{I_{\text{bin}}} \leq I + 1, \quad (109)$$

and are generated by the recursion

$$s_{k+1} = \max(\lfloor r s_k \rfloor, s_k + 1), \quad r > 1, \quad (110)$$

which guaranties at least one integer per bin, even when $r s_k$ floors back to s_k . The bin ratio is determined from a target bin count I_{bin} :

$$r = \max\left((I/i_{\text{discrete}})^{1/I_{\text{bin}}}, 1.01\right). \quad (111)$$

Bin k spans the integer interval $\mathcal{B}_k = \{n \in \mathbb{Z} : s_k \leq n < s_{k+1}\}$ and has width $\Delta_k = s_{k+1} - s_k$. The geometric midpoint of bin k is

$$\hat{s}_k = \sqrt{s_k (s_{k+1} - 1)}. \quad (112)$$

The vacancy axis is partitioned identically with parameters v_{discrete} , V_{bin} , edges $\{t_k\}_{k=0}^{V_{\text{bin}}}$, and ratio $r_v = (V/v_{\text{discrete}})^{1/V_{\text{bin}}}$.

For each bin k of the SIA partition, define

$$\mu_k^{(0)}(t) \equiv \sum_{n \in \mathcal{B}_k} c_n(t), \quad \mu_k^{(1)}(t) \equiv \sum_{n \in \mathcal{B}_k} n c_n(t), \quad \mu_k^{(2)}(t) \equiv \sum_{n \in \mathcal{B}_k} n^2 c_n(t). \quad (113)$$

The mean cluster size in bin k is $\langle n \rangle_k = \mu_k^{(1)} / \mu_k^{(0)}$, and the total SIA content is $S_I = \sum_{n=1}^{i_{\text{discrete}}} n c_n + \sum_{k=1}^{I_{\text{bin}}} \mu_k^{(1)}$. The number of moments per bin $P \in \{1, 2, 3\}$ is selected by the shape function parameter: "constant": track $\mu_k^{(0)}$ only ($P = 1$); "linear": track $(\mu_k^{(0)}, \mu_k^{(1)})$ ($P = 2$); and "lognormal": track $(\mu_k^{(0)}, \mu_k^{(1)}, \mu_k^{(2)})$ ($P = 3$). The analogous bin moments $\nu_k^{(p)} = \sum_{m \in \mathcal{B}_k^v} m^p \bar{c}_m$ are defined on the vacancy partition.

Rather than re-deriving the bin-integrated rates symbolically (which introduces explicit inter-bin flux terms and bin-averaged rate constants), we evolve the moments $\{\mu_k^{(p)}\}$ via a three-step procedure executed at every right-hand-side integration call, as follows:

- R** *Reconstruct* a per-size distribution c_n^{rec} on the full integer grid $n = 1, \dots, I$ by combining the discrete unknowns $\{c_n\}_{n \leq i_{\text{discrete}}}$ with the closure ansatz $c_n[\mu_k^{(0)}, \mu_k^{(1)}, \mu_k^{(2)}]$ for $n \in \mathcal{B}_k$ (Sec. 5.5 below). The same is done for the vacancy axis.
- E** *Evaluate* the right-hand sides of the per-size master equations (83), (85), (86) at every n and every m , using c_n^{rec} and $\bar{c}_m^{\text{rec}}$. All reaction channels (production, coalescence, V–I annihilation, cavity absorption, thermal emission, fixed sinks, trap mutation, re-resolution, He capture / emission) are evaluated *without bin-averaging the rate constants*; the bin averaging enters only through the reconstructed concentrations.
- P** *Project* the per-size rates onto the tracked moments:

$$\frac{d\mu_k^{(p)}}{dt} = \sum_{n \in \mathcal{B}_k} n^p \left(\frac{dc_n}{dt} \right)^{\text{full}}, \quad p = 0, 1, \dots, P-1, \quad (114)$$

and analogously for the vacancy moments. Sizes $n \leq i_{\text{discrete}}$ are copied verbatim into the discrete part of $d\mathbf{y}/dt$.

This is the discrete analog of the Galerkin projection (123) below: step **P** is the test-function integration, with n^p playing the role of $\phi_p(n)$ on the integer grid $\mathcal{B}_k$. Because the per-size rate evaluation is exact, all conservation properties of the master equations (Eqs. 83–86) are inherited automatically: no inter-bin flux $J_{k \rightarrow k+1}$ has to be defined or made upwind-consistent, and no bin-averaged rate constants $\hat{\mathcal{K}}_{k,m}^{(\text{cav})}$ appear. The only approximation is the closure used in step **R**. The fixed-sink term retains its size dependence, $\sum_n \mathcal{D}_i(n) c_n$, because $\mathcal{D}_i(n) \propto \omega_n^{\text{eff}}(n)$ varies with n through the 1D-glide diffusivity.

The vacancy block obeys the analog of the three-step procedure, with $n \rightarrow m$ and the corresponding rate constants from (85), including: V–V coalescence, thermal vacancy emission $\varepsilon_v(m, \ell)$ (with the He-pressure correction from Sec. 5.1), SIA-induced shrinkage for all mobile $n = 1, \dots, n_{\text{max}}^i$, and the fixed-sink loss $\sum_{m \leq m_{\text{max}}^v} \mathcal{D}_v \bar{c}_m$. He capture does *not* change the vacancy size class m , so it appears in the dQ/dt equations only. The two He-reduction modes of Sec. 5.1 are carried over unchanged. In Case 2 (fission, decoupled scalar ℓ_{tot}) a single scalar Q_{tot} is evolved alongside the moments; in Case 1 (fusion, mean-field $\bar{\ell}(m)$) one Q_k is attached to each vacancy bin, so that $\bar{\ell}_k = Q_k / \nu_k^{(0)}$ is constant within bin k .

5.5. Closure of Bin-Moment Equations

The reconstruction step **R** above requires a deterministic map from the tracked moments $(\mu_k^{(0)}, \mu_k^{(1)}, \mu_k^{(2)})$ back to per-size concentrations c_n for $n \in \mathcal{B}_k$. Three closures are implemented.

Piecewise-constant ($P=1$). The simplest closure assumes that the distribution is uniform within each bin:

$$c_n \approx \frac{\mu_k^{(0)}}{\Delta_k} \quad \forall n \in \mathcal{B}_k. \quad (115)$$

This is the classical *logarithmic grouping* [35, 36, 50, 52, 54]. It preserves $\mu_k^{(0)}$ exactly; the first-moment error per bin is $O(\Delta_k / \hat{s}_k) = O(r - 1)$. No first-moment ODE is solved in this mode.

Linear dual basis ($P=2$, “hat-function”). With two tracked moments per bin, choose two basis functions $(\phi_{k,0}(n), \phi_{k,1}(n))$ such that

$$c_n \approx \phi_{k,0}(n) \mu_k^{(0)} + \phi_{k,1}(n) \mu_k^{(1)}, \quad n \in \mathcal{B}_k. \quad (116)$$

The two basis functions are the linear *dual basis* on the discrete index set $\mathcal{B}_k$, i.e. the unique pair satisfying

$$\sum_{n \in \mathcal{B}_k} \phi_{k,0}(n) = 1, \quad \sum_{n \in \mathcal{B}_k} \phi_{k,1}(n) = 0, \quad \sum_{n \in \mathcal{B}_k} n \phi_{k,0}(n) = 0, \quad \sum_{n \in \mathcal{B}_k} n \phi_{k,1}(n) = 1. \quad (117)$$

Solving these constraints for the linear ansatz $\phi_{k,p}(n) = \alpha_p + \beta_p n$ gives the closed form used in the code:

$$\phi_{k,0}(n) = \frac{S_2 - S_1 n}{\mathcal{N}_k}, \quad \phi_{k,1}(n) = \frac{\Delta_k n - S_1}{\mathcal{N}_k}, \quad (118)$$

where the geometry integrals are

$$S_1 \equiv \sum_{n \in \mathcal{B}_k} n, \quad S_2 \equiv \sum_{n \in \mathcal{B}_k} n^2, \quad \mathcal{N}_k \equiv \Delta_k S_2 - S_1^2 > 0, \quad (119)$$

and are computed once at setup. Equations (118) and (119) are evaluated on the *integer* grid $\mathcal{B}_k = \{s_k, s_k + 1, \dots, s_{k+1} - 1\}$; they reduce to piecewise-linear FEM hat functions only in the continuum limit $\Delta_k \rightarrow \infty$. Negative reconstructed values are clamped to zero before use in step **E**; this clamp breaks exact moment preservation only for strongly-skewed bins.

Log-normal closure ($P=3$). Within each bin, assume the per-size concentration follows a log-normal distribution in n :

$$c_n \approx \frac{a_k}{n} \exp\left[-\frac{(\ln n - m_k)^2}{2 \sigma_k^2}\right], \quad n \in \mathcal{B}_k, \quad (120)$$

parameterized by (a_k, m_k, σ_k^2) . Given the three tracked moments, define $\bar{n}_k = \mu_k^{(1)}/\mu_k^{(0)}$ and $\bar{n}_k^2 = \mu_k^{(2)}/\mu_k^{(0)}$. The log-normal shape parameters are obtained from the moment matching equations:

$$\sigma_k^2 = \ln\left(\frac{\mu_k^{(2)} \mu_k^{(0)}}{(\mu_k^{(1)})^2}\right), \quad m_k = \ln \bar{n}_k - \frac{1}{2} \sigma_k^2. \quad (121)$$

The amplitude a_k is fixed by requiring *discrete* preservation of the zeroth moment over $\mathcal{B}_k$:

$$a_k = \frac{\mu_k^{(0)}}{\sum_{n \in \mathcal{B}_k} \frac{1}{n} \exp\left[-\frac{(\ln n - m_k)^2}{2 \sigma_k^2}\right]}. \quad (122)$$

Note: only $\mu_k^{(0)}$ is exactly preserved by this discrete normalization. When $\sigma_k^2 \leq 0$ (monodisperse or non-physical moments) the code falls back to the linear dual basis, and when $\Delta_k = 1$ to the single-size assignment $c_{s_k} = \mu_k^{(0)}$.

Galerkin formulation (general). The three closures above are special cases of the general Galerkin projection of the master equation (83) onto basis functions $\{\phi_p\}_{p=0}^{P-1}$ defined on $\mathcal{B}_k$:

$$\sum_{n \in \mathcal{B}_k} \phi_p(n) \frac{dc_n}{dt} = \sum_{n \in \mathcal{B}_k} \phi_p(n) f_n(\mathbf{c}), \quad p = 0, 1, \dots, P-1, \quad (123)$$

with f_n the right-hand side of (83). Choosing $\phi_p(n) = n^p$ as the test functions yields the moment projection (114) used in the code; the trial functions enter only through the reconstruction step **R**.

Mass conservation. Because the projection (114) sums the *exact* per-size rates, the bin-moment system inherits the Frenkel-pair and helium-balance identities of the full per-size system without modification. In particular, the swelling identity (Sec. 4.5) holds exactly at the level of the moment ODE system, with S_I and S assembled as the discrete + first-moment sums introduced after Eq. (113).

Truncation errors. Closure error enters only through the reconstructed c_n in step **R**. For a smooth, slowly varying distribution $f(n) = c_n$ inside a bin and any reaction kernel $\mathcal{K}(n)$ that is smooth on the same scale:

$$\sum_{n \in \mathcal{B}_k} \mathcal{K}(n) c_n - \sum_{n \in \mathcal{B}_k} \mathcal{K}(n) c_n^{\text{rec}} = \begin{cases} O((r-1)^2), & P = 1 \text{ (piecewise-constant)}, \\ O((r-1)^3), & P = 2 \text{ (linear dual basis)}, \\ O((r-1)^4), & P = 3 \text{ (log-normal, smooth limit)}. \end{cases} \quad (124)$$

For $r = 2$ this gives $\sim 25\%$, $\sim 12\%$, and $\sim 6\%$ per-bin rate-constant error respectively; for $r = 1.5$, $\sim 6\%$, $\sim 1.5\%$, $\sim 0.4\%$. The lognormal closure is rate-limited by the integer-grid residuals in $\mu_k^{(1)}, \mu_k^{(2)}$ (Eq. 122); in practice, the linear dual basis with $r \approx 1.5$ –2 is the best accuracy/cost compromise and is the default.

6. Results

This section showcases three principal computational strengths of RAD-CLUSTER. First, running on a single OpenMP-enabled workstation, the code

is capable of simulating the nucleation and growth of experimentally observable defect clusters at irradiation doses reaching 100 dpa, with wall-clock times ranging from 10 to 200 minutes. Second, the code is used to investigate how fission and fusion neutron spectra differ in their effects on microstructure evolution, and to assess the role of SIA cluster mobility in that evolution. Third, the sensitivity of both simulation run time and key microstructural predictions to the choice of numerical solver is examined. The intent is not to carry out a comprehensive validation against the full body of experimental data, but rather to demonstrate the viability of the modeling framework and to lay the groundwork for subsequent studies in which machine-learning techniques will be employed to systematically calibrate model parameters to experimental observations.

A natural question arises: given the high-fidelity nature of atomistic data, can one reproduce **all** experimentally observable features of the irradiated microstructure in a complex alloy such as EUROFER-97? If the “ground truth” is assumed to be represented by TEM observations on the experimental side and by DFT/MD defect data on the modeling side, an important issue then becomes the sensitivity of the results to uncertainties inherent in the underlying database. Addressing this problem of uncertainty quantification is nontrivial and warrants a dedicated investigation using modern AI-based methodologies.

Experimental observations clearly show that the most dramatic influence on both loop density and mean size is the irradiation temperature. Figure 3 illustrates the characteristic non-monotonic behavior schematically, supported by data from EUROFER-97 (16.3 dpa, BOR-60) [55], T91 (15.4 dpa, BOR-60) [56], and T91 dual-ion (35 dpa) [57]. All simulations presented here were performed at 300 °C using the input parameters listed in Tables 2 and 3 and the target experimentally observed cluster mean size and density at that temperature shown in Figure 3.

The production fractions presented in Table 2 are *reconstructed* from published MD cascade data [9, 18, 20, 58–69]. The fundamental kinetic variables are the EUROFER-97 effective reaction frequencies

$$\omega_{\alpha}^{\text{eff}} = \frac{\nu_{\alpha} \exp(-E_m^{\alpha}/k_B T)}{1 + \sum_s z_s c_s \exp(E_b^{\alpha-s}/k_B T)}, \quad \alpha \in \{i, v, h\}, \quad (125)$$

which incorporate solute trapping by Cr, W, and minor-alloying solutes through trap concentrations c_s and binding energies $E_b^{\alpha-s}$. A list of material and defect energetics data is shown in Table 3, obtained from references

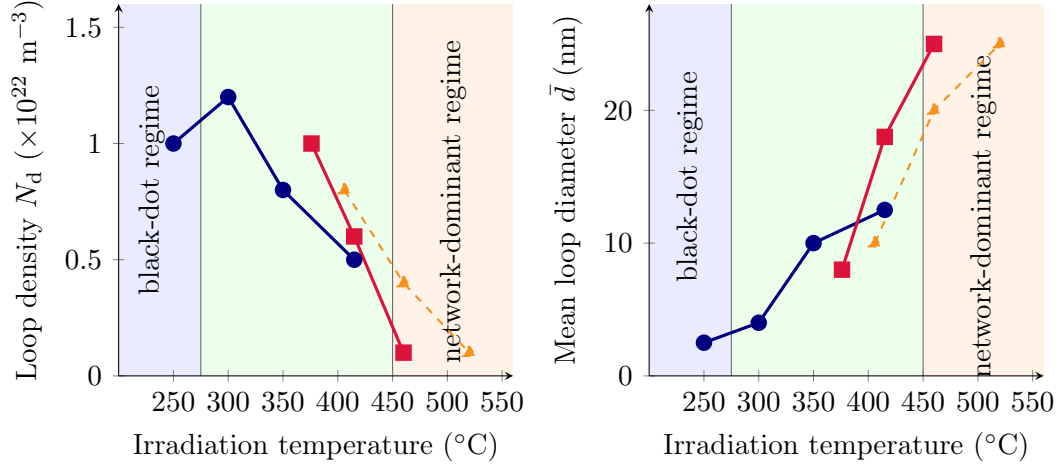


Figure 3: Temperature dependence of (a) loop number density N_d and (b) mean loop diameter $\bar{d}$ at fixed dose. In both panels, blue circles denote EUROFER-97 at 16.3 dpa [55], red squares denote T91 at 15.4 dpa under BOR-60 neutron irradiation [56], and orange dashed triangles denote T91 under dual-ion irradiation at 35 dpa [57]. The peak loop density in EUROFER-97 occurs at approximately 300 $^{\circ}\text{C}$. The ion-irradiated T91 data are shifted by approximately 60–70 $^{\circ}\text{C}$ relative to the neutron data [57].

[21, 45, 70–81]. A full quantification of modeling and experimental uncertainties in the database is in the planning stage and is outside the scope of the current investigation.

Table 2: Comparison of defect production characteristics under fission and fusion neutron spectra in α -Fe [9, 18, 20, 58–69].

Quantity	Fission	Fusion
Mono-SIA fraction, $1 - f_i^{\text{cl}}$	0.42	0.35
Mono-vacancy fraction, $1 - f_v^{\text{cl}}$	0.85	0.80
Mean SIA cluster size $\langle m \rangle_{\text{cl}}$	4.4	6.3
Mean vacancy cluster size $\langle n \rangle_{\text{cl}}$	2.6	3.1
Largest SIA cluster i_{cascade}	20	50
Largest vacancy cluster v_{cascade}	10	20
He co-production (appm He/dpa)	0.5–1	~ 10

Table 3: Reaction rate input parameters for EUROFER cluster dynamics [21, 45, 70–81].

Parameter	Symbol	Value	Units
<i>Dislocation sink:</i>			
Network dislocation density	ρ_d	10^{13} – 10^{14}	m^{-2}
Interstitial bias factor	Z_i^d	1.02–1.15	—
<i>Grain-boundary sink:</i>			
Grain diameter	d_g	5–20	μm
<i>Precipitate sink:</i>			
Mean precipitate radius	r_p	10–50	nm
Precipitate number density	N_p	10^{20} – 10^{21}	m^{-3}
<i>Mobility cutoffs:</i>			
Max mobile SIA cluster	i_{mobile}	100	SIAs
Max mobile vacancy cluster	v_{mobile}	5	vacancies
<i>Vacancy energetics:</i>			
Formation energy	E_f^v	2.0	eV
Migration energy	E_m^v	0.67	eV
Divacancy binding energy (2NN)	E_B^{2v}	0.22	eV
Divacancy migration energy	E_m^{2v}	0.70	eV
<i>SIA energetics:</i>			
Formation energy ($\langle 110 \rangle$ dumbbell)	E_f^i	3.64	eV
Migration energy ($\langle 110 \rangle$ T-R)	E_m^i	0.34	eV
Di-SIA binding energy	E_B^{2i}	0.80	eV
Di-SIA migration energy	E_m^{2i}	0.42	eV
<i>Helium energetics:</i>			
He migration energy (tetrahedral)	E_m^h	0.06	eV
He–vacancy binding (1st He)	E_b^{h-V}	2.30	eV
He–vacancy binding (2nd He)	$E_b^{h-V_2}$	2.00	eV
<i>Attempt frequencies:</i>			
Vacancy	ν_v	1.0×10^{13}	s^{-1}
SIA ($\langle 110 \rangle$)	ν_i	1.0×10^{13}	s^{-1}
He interstitial	ν_h	3.0×10^{12}	s^{-1}
<i>He equation of state (virial coefficients, 600–1000 K):</i>			
Second virial coefficient	B_2	1.67×10^{-29}	m^3/atom
Third virial coefficient	B_3	1.84×10^{-58}	m^6/atom^2

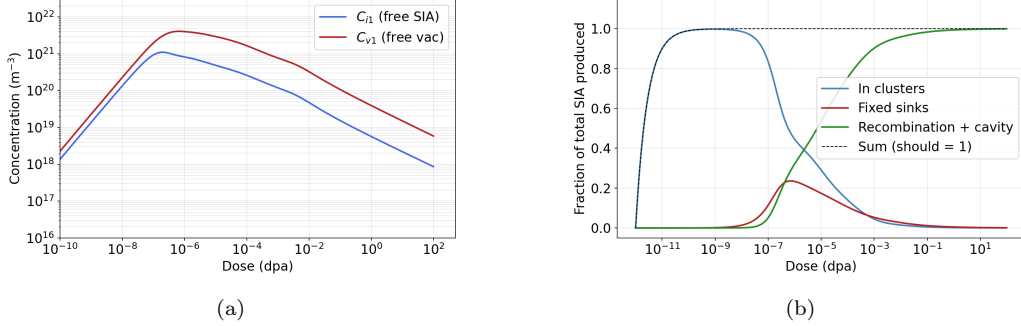


Figure 4: (4a) Evolution of point defect concentrations in EUROFER-97. Notice the long nucleation transient due to the continuous formation of defect clusters. (4b) Evolution of the partitioned fractions of SIA clusters. Note that the majority are absorbed at sinks or have recombined with vacancy clusters.

6.1. High Dose Simulation Results

Figure 4a shows the evolution of the mono-vacancy and mono-SIA concentrations in EUROFER-97. The long nucleation transient is a direct consequence of the continuous coalescence of highly mobile SIA clusters: small interstitial clusters in BCC iron migrate one-dimensionally with very low activation energies and continually feed the cluster population through cluster-cluster reactions, a channel that treatment restricted to mono-defect diffusion cannot reproduce. Figure 4b shows the components that contribute to the mass conservation of SIA clusters, indicating that the majority are lost to absorption at dislocations and grain boundaries or to mutual recombination with vacancy clusters. These two channels control the net defect retention available for hardening and swelling and explain why the cluster population eventually saturates.

Figure 5a tracks the dose evolution of the SIA and vacancy cluster number densities, exhibiting the incubation, rapid nucleation, and saturation stages. The predicted densities lie within the same order of magnitude as TEM measurements on EUROFER-97 [82], even though the input parameter space (formation and binding energies, sink strengths, bias factors, and cascade source terms) was not optimized. This indicates that the model captures the dominant physics of cluster mobility, sink bias, and recombination. Figure 5b complements this by resolving the cavity concentration binned by size: small vacancy clusters nucleate first and progressively feed larger cavities as the master-equation kinetics set the population of each size bin

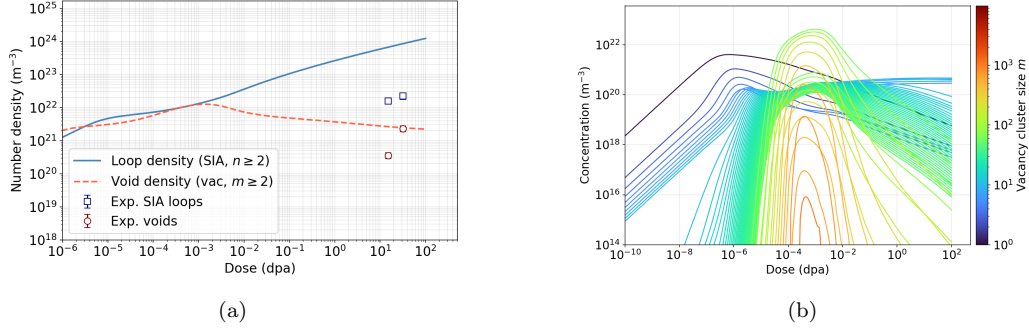


Figure 5: (5a) Evolution of defect cluster density in EUROFER-97. (5b) Evolution of the concentration of various cavity size as function of dose. Experimental data are obtained from reference [82].

through the competition between vacancy capture, SIA-induced shrinkage, and thermal emission.

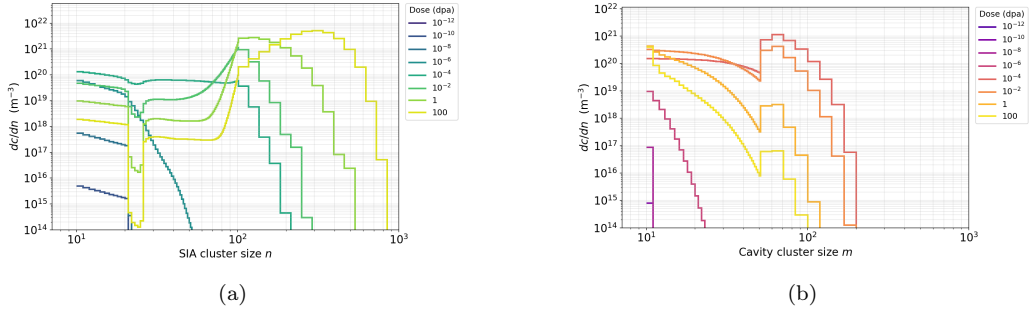


Figure 6: (6a) Evolution of SIA defect cluster size distribution in EUROFER-97. (6b) Evolution of vacancy cluster defect cluster size distribution in EUROFER-97.

The size distributions in Figures 6a and 6b reveal the asymmetry between SIA and vacancy kinetics. The SIA distribution is comparatively narrow and shifts rapidly toward larger sizes, a signature of the strong coalescence channel enabled by 1D-glissile motion: once nucleated, loops grow predominantly by absorbing mobile clusters rather than by single-interstitial capture. The vacancy distribution, by contrast, is broader and develops more slowly, reflecting the much lower mobility of vacancy clusters and the central role of thermal emission in shaping the high-size tail. That a single workstation can resolve these distributions across many decades in size and time by solving more than 20,000 coupled stiff ODEs is itself a notable result, demonstrating that mesoscale defect evolution in a complex alloy like EUROFER-97

is tractable and opens the door to future potential couplings with machine learning, spatial dependence, dislocation dynamics, multiscale physics, etc.

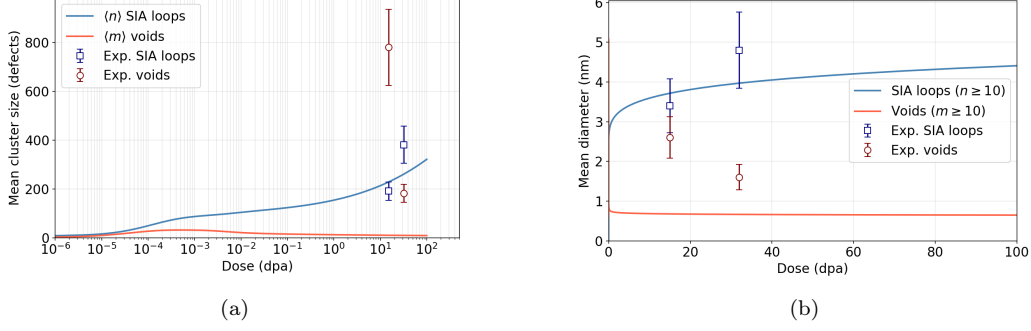


Figure 7: (7a) Evolution of the mean number of atoms (vacancies) of SIA & vacancy defect clusters compared to experiments. (7b) Evolution of SIA and vacancy cluster defect cluster mean size compared to experiments. Experimental data are obtained from reference [82].

The mean SIA loop and cavity sizes simulated here are in reasonable agreement with TEM observations on EUROFER-97 across the explored dose range, especially considering that the input parameters were not tuned to the experiment [82]. The mean SIA size grows monotonically with dose as a balance of cluster–cluster coalescence and absorption at sinks, while the mean cavity size evolves more slowly under the combined action of net vacancy flux, dislocation bias, and thermal emission from small cavities. Together with the saturating densities of Figure 5, these results indicate that a generalized cluster dynamics treatment can quantitatively reproduce the irradiation response of a technologically complex reduced-activation ferritic–martensitic steel without recourse to fitted parameters.

6.2. Phase Space Reduction and Solver Configuration Studies

The high-fidelity simulations presented above involve more than 20,000 coupled stiff ODEs, which makes them computationally demanding and motivates a systematic assessment of phase-space reduction methods and solver configurations. Two questions must be addressed before such reduced models can be deployed in production: first, which physical assumptions (helium-kinetics treatment, cascade spectrum, mobility cutoffs, sink-density level) can be relaxed or simplified without distorting the macroscopic observables; and second, which numerical strategies offered by RADCLUSTER remain simultaneously accurate, stable, and tractable at the scales of interest. RADCLUSTER supports several solver back-ends and discretization strategies, namely

a fully discrete formulation solved with either GMRES+Woodbury preconditioning (suitable when the dense rank-one coupling from coalescence dominates) or GMRES+Jacobi preconditioning (lighter but only effective when the system is weakly coupled), a sliding-window solver (applicable when the active region of phase space migrates monotonically toward larger sizes), and a bin-moment reduction that closes the high-size tail with a piecewise-constant ($P = 1$), linear-hat ($P = 2$), or lognormal ($P = 3$) ansatz, applicable whenever the underlying distribution is smooth on the scale of a bin. The studies below probe each of these knobs at the 10 dpa horizon in order to identify the configuration that delivers reference-quality results at the lowest cost.

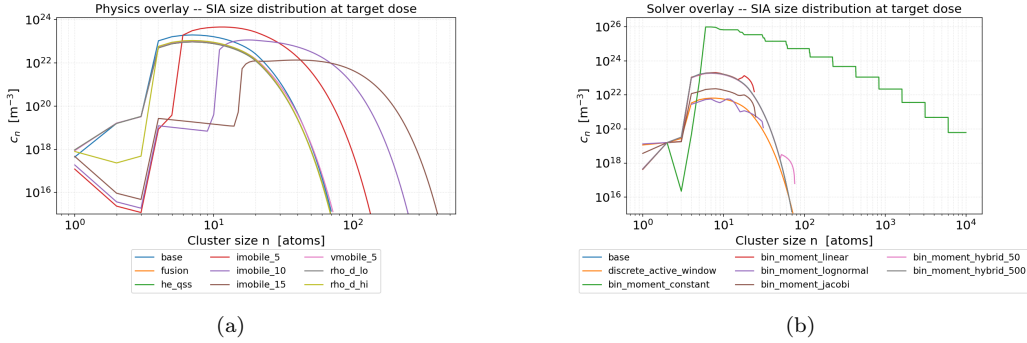


Figure 8: (8a) The influence of physical parameters on SIA defect cluster size distributions at 0.1 dpa in EUROFER-97. (8b) The effects of numerical solver configuration on SIA cluster defect cluster size distributions at 0.1 dpa in EUROFER-97.

As shown in Fig. 8a, the He-kinetics treatment (dynamic vs. quasi-steady-state) and the cascade spectrum (fission vs. fusion) leave the SIA and vacancy size distributions, the loop number density $N_{\text{loops}} \simeq 9.8 \times 10^{23} \text{ m}^{-3}$, and the mean cluster sizes $\langle n_i \rangle \simeq 10.4$, $\langle m_v \rangle \simeq 845$ essentially indistinguishable from the mono-defect baseline; only the helium inventory diverges sharply, with the fusion spectrum trapping $C_{\text{He}}^{\text{tot}} = 6.6 \times 10^{23} \text{ m}^{-3}$ versus $2.8 \times 10^{20} \text{ m}^{-3}$ for fission — a $\sim 2000\times$ separation that grows by two further decades between 0.1 and 10 dpa as the fusion-rate G_{He}/G accumulates. The SIA-mobility cutoff i_{mobile} is the dominant phase-space knob, and its effect amplifies with dose: at $i_{\text{mobile}}=5$, the loop density rises to $7.6 \times 10^{24} \text{ m}^{-3}$ (a factor of eight above mono-defect) with $\langle n_i \rangle$ shifting from 10.4 to 16.8, while $i_{\text{mobile}}=10$ already pushes the system into a stiffer regime ($N_{\text{loops}} = 3.5 \times 10^{24}$, $\langle n_i \rangle = 32.6$, $\langle m_v \rangle = 196$). Vacancy-cluster mobility ($v_{\text{mobile}}=5$) perturbs the loop

and void populations by only $\sim 10\%$, and a decade increase in the dislocation sink density ($\rho_d = 10^{15} \text{ m}^{-2}$) reduces N_{voids} by $\sim 12\%$ and free helium by two orders of magnitude.

Figure 8b reports the corresponding solver comparison study. At the domain of $I = V = 10,000$ ($N_{\text{eq}} = 20,007$ for the discrete formulation), only two configurations actually reach 10 dpa within 10 000 s of wall-clock time on a single core of an Intel Xeon Scalable processor (Skylake-SP/Cascade Lake-SP, Family 6 Model 85, with AVX-512 support) workstation with 256 GB of RAM: the GMRES+Woodbury discrete reference (9198 s; $N_{\text{loops}} = 2.04 \times 10^{24} \text{ m}^{-3}$, $\langle n_i \rangle = 10.4$, $\langle m_v \rangle = 4131$) and the bin-moment reduction with the linear hat closure ($P = 2$, $N_{\text{eq}} = 65$), which completes in 213 s — a $\sim 43\times$ speed-up while reproducing every macroscopic target within $\lesssim 3\%$ of the reference ($N_{\text{loops}} = 2.09 \times 10^{24}$, $\langle m_v \rangle = 4058$). All other variants fail before 10 dpa: the piecewise-constant closure ($P = 1$) inflates N_{loops} by three orders of magnitude ($2.4 \times 10^{27} \text{ m}^{-3}$) and the mean SIA size by a factor of eight, confirming that the linear hat is the minimal closure that preserves moments correctly. The lognormal closure ($P = 3$), the GMRES+Jacobi variant, and the sliding-window solver time out. This solver comparison indicates that the bin-moment linear-closure/GMRES+Woodbury combination is the best solver configuration that is simultaneously accurate and computationally tractable for routine production runs.

7. Summary & Conclusions

This paper has developed a general, modular, graph-based cluster-dynamics (CD) framework for the long-term evolution of radiation-induced defect populations in crystalline solids and has demonstrated it on the ferritic–martensitic steel EUROFER-97 under prototypical fission and fusion irradiation. The presentation proceeds, in a unified notation, from the reaction admissibility graph and its master equations to the helium–vacancy state-space reductions, the logarithmic size-bin moment closures, the tailored numerical solver, and demonstration runs covering 10^{-6} to 100 dpa.

Principal contributions. The first contribution is a pair of algorithmic primitives that make a generic CD formulation possible. The *polarity index* $\sigma \in \{-1, +1\}$ separates the cluster state space into vacancy and SIA-type sub-spaces without hard-coding a host-specific taxonomy, while the *reaction admissibility graph* (RAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ encodes every elementary reac-

tion—growth, shrinkage, dissociation, recombination, solute trapping, inter-population transfer, coalescence, source, and sink—as a labeled edge between admissible vertices. Physically forbidden reactions are excluded during graph construction, leaving the master equation as the host-independent graph-walking accumulator $dc_a/dt = G_a + \sum_r S_{ar}J_r(\mathbf{c}) - D_a c_a$ applied edge-by-edge over $\mathcal{E}$.

The second contribution is the specialization of this machinery to ferritic–martensitic steels: Section 4 works out the full RAG for EUROFER-97 together with size-resolved equations for SIA clusters, vacancy/helium cavities, and free helium in both general-coalescence and mono-defect forms. The Frenkel-pair and helium-conservation laws follow directly from the stoichiometric matrix and hold exactly at the continuous level. Two helium-occupancy reductions emerge as systematic limits of the same parent equations: a mean-field loading $\bar{\ell}(m)$ per cavity size for fusion-relevant He/dpa ratios, and a single decoupled inventory Q_{tot} for fission spectra; both preserve the conservation moments by construction.

The third contribution is the logarithmic size-bin moment reduction. Evolving the zeroth and first moments on a geometric size grid cuts the cluster-size ODE count by roughly an order of magnitude at controlled accuracy. A piecewise-constant closure—the minimal one consistent with both moments—and a linear-hat Galerkin closure are developed and analyzed; both converge to the discrete reference as $P \rightarrow 1$ and conserve mass to machine precision when the boundary flux is handled consistently between adjacent bins.

The fourth contribution is the production-grade numerical core. The CD Jacobian is bordered–banded—narrow-banded in the size index within each species and bordered through cross-species couplings (sink strengths, helium balance, dislocation density). A tailored Sherman–Morrison–Woodbury preconditioner splits the dense border updates from the banded core and supports efficient GMRES solves across every dose-rate decade, while in-line conservation diagnostics provide continuous quality control. The entire formulation is shipped in the open-source RADCLUSTER code, which exposes both the reaction graph and the numerical knobs as configurable inputs so that one source tree covers fission, fusion, and ion-beam scenarios.

Lessons from the simulations. The runs in Section 6 confirm that the framework produces quantitatively sensible defect populations for EUROFER-97 across the explored dose envelope, with the linear-hat bin-moment reduction

reproducing every macroscopic target of the fully discrete reference within a few percent and delivering a $\sim 40\times$ wall-clock speed-up. The lognormal and sliding-window variants either inflate the loop density by orders of magnitude or fail before 10 dpa, leaving the linear-hat/GMRES+Woodbury combination as the recommended production configuration: simultaneously accurate, conservative, and tractable. Mean SIA loop and cavity sizes match TEM observations on EUROFER-97 without parameter tuning, indicating that the input physics is internally consistent.

The parametric study identifies the SIA-mobility cutoff i_{mobile} as the dominant phase-space knob: shifting $i_{\text{mobile}} = 1 \rightarrow 5$ multiplies the loop density by nearly an order of magnitude and the mean loop size by a comparable factor, with effects that amplify with dose. Vacancy-cluster mobility perturbs the same populations by only $\sim 10\%$, and a decade increase in dislocation sink density reduces void densities by $\sim 12\%$ and free-helium content by two orders of magnitude — rankings consistent with sink-strength arguments and a reminder that host-specific identification of i_{mobile} is essential before quantitative predictions.

Outlook. Several directions are immediately open. The RAG already accommodates additional species (hydrogen and other transmutation products, $\langle a \rangle/\langle c \rangle$ loops in HCP hosts, carbide, or α' precipitates) by extending the vertex set and admissibility filter without touching the numerical kernel. The bin-moment closures generalize to multi-dimensional cluster spaces—size with composition or helium content—once the boundary-flux bookkeeping is extended along each axis, opening the natural route to coupled solute segregation and precipitation. The bordered-banded structure is preserved under spatial discretization, so embedding the same kernel in a 1D reactor-fuel or first-wall transport calculation, or coupling it to discrete dislocation dynamics or phase-field microstructure models, is a straightforward extension rather than a rewrite. Finally, the modular split between the reaction graph and its rate kernels allows data-driven surrogates—neural closures for binding energies, learned coarse-grained rate constants, or active-learning loops over atomistic calibrations—to replace individual edge labels without disturbing the conservation laws or the solver. Together, these features position the framework as a more reliable bridge between atomistic calibration and engineering-scale prediction of irradiated materials.

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Declarations

- **Ethics Declaration:** The author declares that this work is original, has been conducted in accordance with accepted standards of research integrity, and does not involve human or animal subjects.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Competing interests:** The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Author contributions:** Nasr M. Ghoniem conceived the work, performed the analysis, developed the code, and wrote the manuscript.
- **Declaration of generative AI-assisted technologies :** During the preparation of this work, the author(s) used Anthropic Claude in order to research published literature, assist in coding, and prepare figures. After using this service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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